

Basics of Agriculture

HuNu2073



**BETTER
PRODUCTION**



**BETTER
NUTRITION**



**BETTER
ENVIRONMENT**



**BETTER
LIFE**



1. NATURAL RESOURCES MANAGEMENT

“Caring for Earth is caring for our self and our community” Daniel Christian Wahl

“Healing the land is healing ourselves.” Adrianna Acosta

“Nature and biodiversity are rising up the business agenda faster than anything else”. Steve Kenzie from
the UN Global Compact Network

1.1. Introduction to Natural Resources

- A resource is a form of energy and or matter which is essential for the functioning of living organisms (population and ecosystem). In other words, any part of the environment such as land, water, air, minerals, forest, wild life etc. that the man can utilize to promote his welfare may be regarded as resources.
- A natural capital /resource/ is any asset that can be obtained from the environment e.g. water, soil, plants, wind, animals, minerals, the energy of the sun and many others.

1.1.1. Classification of Natural Resources

A prominent way to classify natural resources is to look in terms of depletion risk.

- A renewable natural resource is one that can regrow, or whose supplies can be replenished through natural processes. E.g. plants, animals, insects, or wind.
- A non-renewable natural resource is one that can be used up or that will not replenish itself. E.g. oil, coal, minerals, or rocks. The use of these resources should be carefully monitored and managed according to their availability. The effects that their exploitation has on all the other resources and the environment as a whole should be carefully followed.

1.1.2. Ecosystem

What is an ecosystem?

What is an Ecosystem Structure?

The **structure** of an **ecosystem** is essentially a description of an environment's organisms and physical features.

It is described under two major headings: Abiotic components and Biotic components.

Ecosystem services

- Ecosystem services are the direct and indirect contributions of ecosystems to human well-being. Healthy, bio-diverse ecosystems sustain humanity by providing air, water and other essential services.
- Many (ecosystem) goods and services e.g. wildlife habitat and diversity, watershed services, carbon storage, and scenic landscapes - are traditionally viewed as free benefits to society, or 'public goods'. Lacking a formal market, these natural assets are traditionally absent from society's balance sheet; their critical contributions are often overlooked in public, corporate, and individual decision-making.
- The degradation of ecosystems has major consequences for our own health and wellbeing. Healthy ecosystems provide us with a wealth of ecosystem goods and services such as water purification, crop pollination, flood protection, **carbon storage and sequestration**. Their monetary value is considerable and virtually irreplaceable.

...Ecosystem services



1.1.3. Over-exploitation

- Being renewable does not mean that these resources automatically last forever. If the renewable resources in a particular area are overexploited for a long period of time, it is entirely possible that they may become endangered or even disappear altogether.
- Over-exploitation of natural resources compromises the availability and fair distribution of resources and their associated benefits to the health of ecosystems and the livelihoods and wellbeing of people.

1.2. Soils

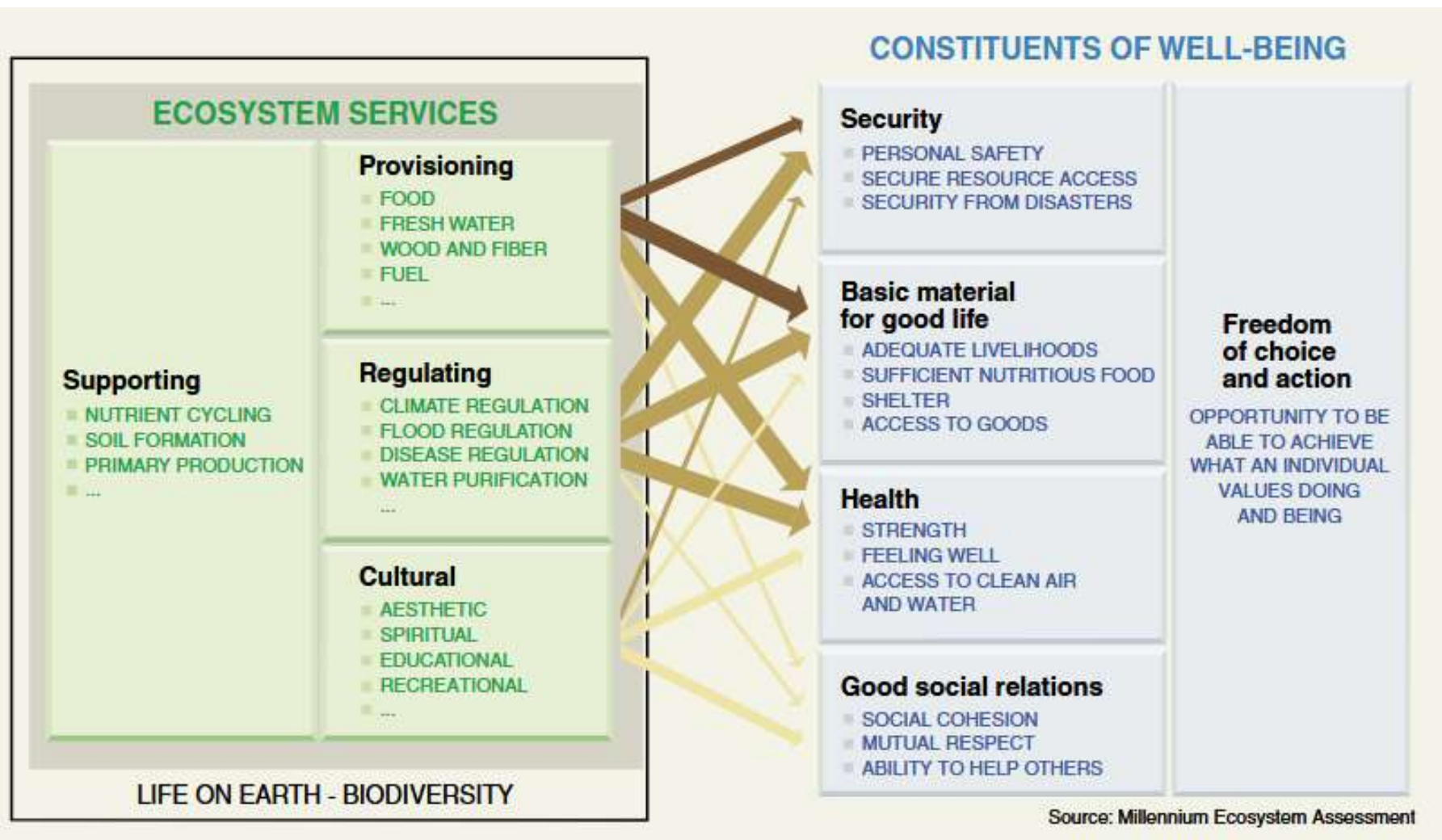


- Soils are fundamental to life on earth, and the ability of soils to provide safe and nutritious food is a key contribution to us and to nature as a whole.
- Soils are the foundation of our food security, providing 95% of the food we eat.
- Food starts with soils. About 95 percent of our food nutrients come from soils, which have a natural capacity to provide nutrients to support crop growth.
- Soil underpins the whole production system.

...1.2. Soils

- Probably, the most recent and appropriate definition of soil is:
A generally loose mineral and/or organic material at or near a planetary surface and usually holds liquids, gases, and biota and support plants.
- Soil forms so slowly that we should regard it as a non-renewable resource.
- Healthy soil , full of biological life, implies that all aspects of the soil, chemical, physical and biological are in a perfect status to be able to produce a healthy food. All people depend on nature and ecosystem services to provide the conditions for a decent, healthy, and secure life (Figure 1). Soil functions are at the core of virtually all processes. **Soil Ecosystem services:** Soil provides, regulates and supports a number of ecosystem services including cultural values.
- Soils are in danger of being the 'orphan' asset, forgotten when food security or forests are discussed. The global decline in soil

Figure 1. Linkages between Ecosystem Services and Human Well-being



...1.2.1. Major Components of Soil

Therefore, mineral soils consist of four major components. These are:

- Mineral (inorganic) constituent: extremely variable in size ranging from 0.0002 to 2 mm in diameter.
- Organic matter: represents an accumulation of partially decayed and partially synthesized plant and animal residues.
- Soil water: makes up soil solution, together with its dissolved salts, which is important as a medium for supplying nutrients to growing plants.
- Soil air: The amount and composition of soil air is determined to a large degree by the water content of the soil.

... 1.2.1. **Major** Components of Soil

Figure 2. Volume composition of a favorable /an ideal/ soil for plant growth

"Does healthy soil smell different?"



... 1.2.2. Soil formation

Soils are formed from rocks by weathering. The major parent material for soils are rocks. Soil formation takes place after a gradual exposure of rocks to the elements and activity of organism – called weathering.

Weathering includes

- physical weathering/disintegration - breaking or grinding particles to smaller sizes without changing their chemical composition.
- chemical weathering/decomposition - the dissolving or chemical alteration of minerals

1.2.2. Soil formation



Many types of plants have adapted to grow on rocks or new soil.

1.2.2.1. Agents

in

Physical Weathering

- Temperature fluctuations
- water/ice
- wind,
- gravity
- mechanical actions of animals, and
- mechanical forces of growing plants

Chemical Weathering

- **Oxygen**: in oxidation-reduction reactions
- **water**: in hydrolysis, hydration and (dis)solution
- **carbon dioxide**: in carbonation
- **organic and inorganic acids**
- **bases**, and
- **salts**

1.2.2.2. Processes in Soil Chemical Weathering

Hydrolysis: is the reaction between the H^+ and/or OH^- ions of water and the elements or ions of a mineral.

Oxidation-reduction reactions: Another chemical process involves the simultaneous loss and gain of electrons in substances.

Hydration: is the binding of water molecules to a mineral.

Carbonation The majority of soil organisms release carbon dioxide (CO_2) from respiration. Carbonation is a reaction by which water dissolves carbon dioxide and the dissolved carbon dioxide forms carbonic acid.

1.2.2.3. Factors of Soil Formation/Soil forming factors

The five soil-forming factors that influence the development of soil are:

- Parent Material
- Topography or position in the landscape
- Climate
- Living organisms, especially vegetation
- Time

1.3. Soils and Plants

In agriculture, using manure/fertilizer returns some organic matter to the soil. Any plant, even annual or perennial weeds, drop leaves and twigs into the soil, which becomes food for soil biota. At the same time, their shade reduces solar radiation from killing microorganisms. Therefore, they play a key role in improving soil fertility and making soil conducive for soil biota to thrive.

1.3.1. Essential Plant Nutrients

The criteria to consider an element to be essential are:

- A deficiency of an essential nutrient makes it impossible for the plant to complete the vegetative or reproductive stage of its life cycle.
- Such deficiency is specific to the element in question and can be prevented or corrected only by supplying this element.
- The element is involved directly in the metabolism of the plant.

...1.3.1. Essential Plant Nutrients

- Sixteen elements are known to be essential in plant growth. They can be grouped based on the amount they are required by plants and based on their sources. Eighteen nutrients are necessary for human health.

Macronutrients: nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulfur (S).

Micronutrients: boron (B), zinc (Zn), manganese (Mn), iron (Fe), copper (Cu), molybdenum (Mo), chlorine (Cl).

- Though the majority of the elements come from the soil, they make up very small part of the plant tissue as plant tissues are mainly made of C, H and O (94-99.5%). C, H and O are not limiting under normal condition as they are supplied by the atmosphere and H₂O.

...1.3.1. Essential Plant Nutrients

A healthy soil is a chemically balanced soil. The macronutrient essential soil nutrients, very important for plant growth, are.

1.3.1.1. Primary Nutrients

🌿 Nitrogen is the major nutrient for strong, vigorous growth, dark green leaf color, and photosynthesis.

🌱 Phosphorus helps roots for strong root growth and development.

🌸 Potassium helps flowers and fruits to develop, improving the overall health and strength of the plant.

1.3.1.2. Secondary Nutrients

Calcium is important for general plant vigor.

Magnesium helps regulate the uptake of other plant foods.

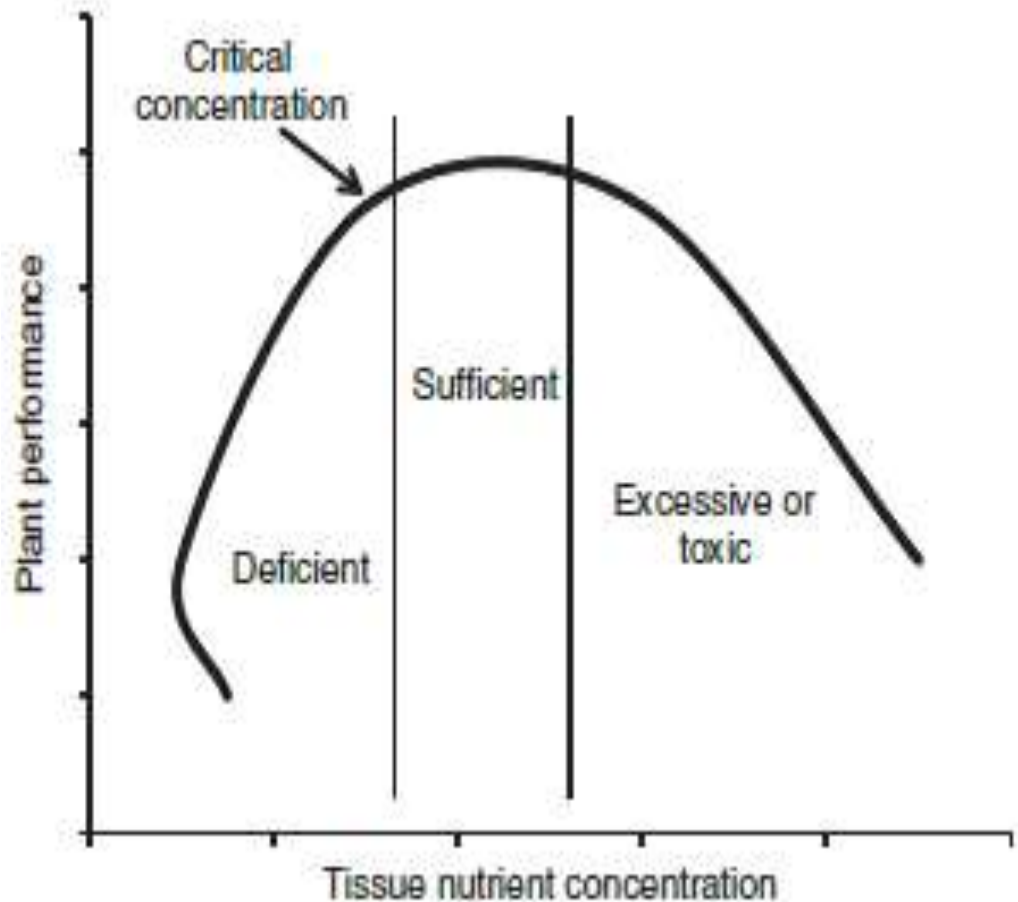
Sulfur helps maintain a dark green color while encouraging more vigorous plant growth.

...1.3.1. Essential Plant Nutrients

- Five additional elements – sodium (Na), cobalt (Co), vanadium (Va), nickel (Ni) and silicon (Si) are essential elements for some plants. Plants absorb many non-essential elements. The content of the micronutrients is also often far in excess of physiological requirements.
- Nutrients are desirable in particular concentrations and can be detrimental to plant growth when concentrations in soil solution are either too low (limiting) or too high (toxicity) [Figure 3].
- The lack of essential nutrients, including macro and micronutrients, leads to the underdevelopment of plants and a decrease in yield and in crop nutritional value.

Nutrient Uptake by Plants

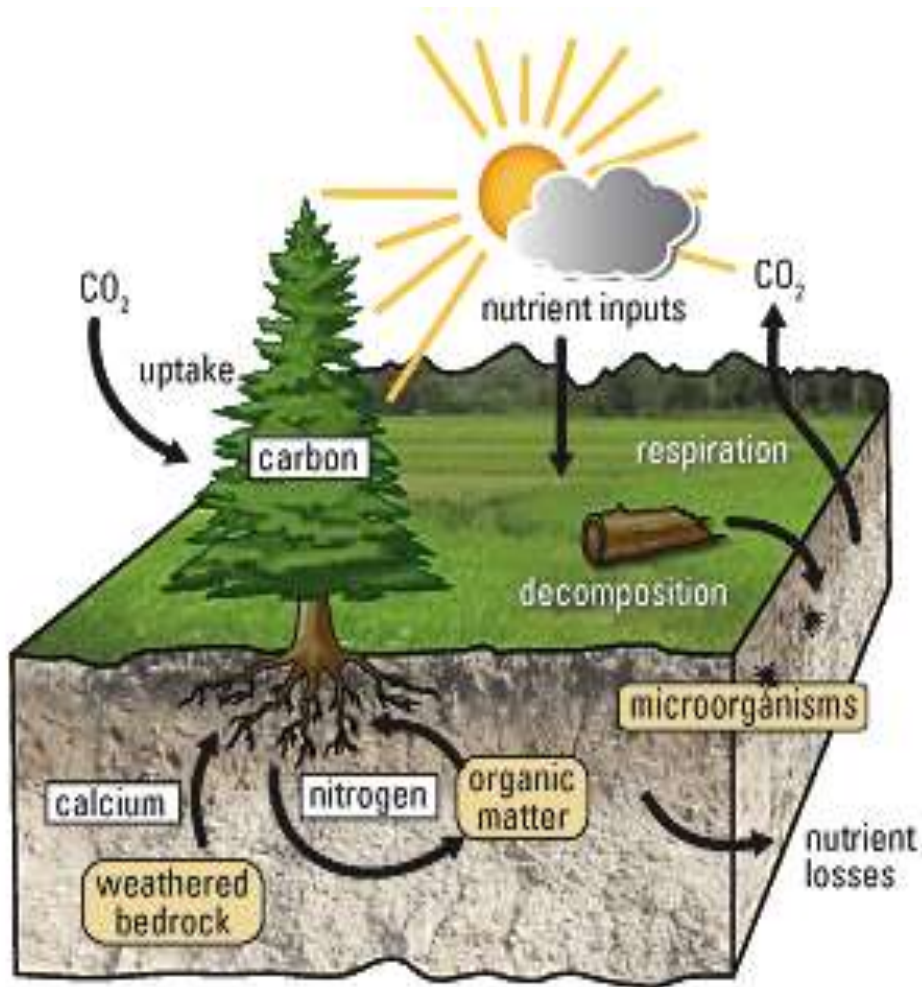
Figure 3. Nutrient concentration in plant tissues as it relates to plant performance. Below the critical range plants are considered deficient. The range where nutrient concentration in the tissue increases, but performance does not, is considered the sufficient range. At some point beyond the sufficiency range, excessive nutrient concentrations can limit plant performance and this is considered the excessive or toxic range.



1.3.2. Soil-Plant Relations

- Beneficial aspects of soils to plants include providing physical support, heat, water, nutrients, and oxygen. Heat, light, and oxygen are also obtained by the atmosphere, but the roots of many plants also require oxygen. Nutrients, dissolved in soil water solution, are derived from soil minerals (inorganic) and organic material (see Figure 3).
- Though many aspects of soil are beneficial to plants, excessively high levels of trace metals (either naturally occurring or anthropogenically added) or applied herbicides can be toxic to some plants.
- The ratio of solids/water/air in soil is critically important to plants for proper oxygen levels and water availability. Too much porosity with air space, such as in sandy or gravelly soils, can lead to less available water to plants, especially during dry seasons when the water table is low. Too much water, in poorly drained regions, can lead to anoxic conditions in the soil, which may be toxic to almost all plants.

Figure 4. Soil-Plant Nutrient Cycle



1.4. Key Pillars of Resource Management

The **main parts** of natural resource management are:

- (1) Forest management/(Agro)forestry
- (2) Water resource management i.e. irrigation and water conservation/harvesting
- (3) Soil resource management and
- (4) Agriculture resource management

Forestry – which yields a vast range of products and ecosystem services useful both for local communities and at a global scale – could play a key role in accelerating a transformation towards societies that simultaneously conserve nature, better provide for human well-being, and generate income, particularly for rural people.

Forestry products can be:

- wood/timber and
- non-timber /non-wood/ forest products (NTFPs/NWFPs)]. Forests create clean air, fertile soils and rich biodiversity.

Agroforestry, an ecologically based traditional farming practice, is the use of trees and shrubs as part of agricultural systems. Planting a range of trees, perennial plants and crops to ensure vegetation stays in the ground year-round.

Sustainable management of forest and agroecosystems requires the maintenance of the supporting natural resources and ecosystem services.

Water resource management mainly includes water supply management and excess water management.

1.5. NR Degradation & Conservation

1.5.1. NR /Environmental/ Degradation

- Land degradation is the progressive reduction in the productive potential of the land to sustain life and provide food security. It is a natural and human-induced phenomenon.
- The world's land area is now 40% degraded and accelerating. This means that greater risk to the climate, biodiversity and the world's ability to feed a growing population. Land is considered 'degraded' when most of its natural resources, soil, water, biodiversity, trees or native vegetation is depleted, many of them which include "green" areas that were intensely farmed or deforested.
- It can be classified under four headings as:
 - 1) Soil degradation
 - 2) desertification
 - 3) vegetation cover degradation
 - 4) water resources degradation

1.5.1.1. Types of Soil Degradation

- Soil degradation is the process which lowers the current and/or future capacity of the soils to produce goods or services. It includes:
 1. Soil erosion: is the displacement of soil from one place to the other. Topsoil is particularly affected and fertility losses are high in relation to physical losses.
 - Water erosion: This is the most widespread soil degradation type in SSA.
 - Wind erosion: Wind erosion occurs most frequently in the arid and semi-arid parts of SSA. If the climate becomes drier or windier, wind erosion is likely to increase.

... 1.5.1.1. Types of Soil Degradation

2. Soil chemical degradation includes processes such as salinity /salinization/ and sodification, acidification, nutrient mismanagement, leaching and soil pollution /build-up of toxicity/.

Agricultural systems lose nutrients with each harvest, and if soils are not fertilized, fertility is progressively lost, and soils will produce nutrient-deficient plants. Nutrient depletion (gradual loss of fertility) occurs if agriculture is practiced on poor or moderately fertile soils, without sufficient application of organic or chemical fertilizers.

Nutrient imbalance is a significant obstacle on the road to food security as it directly affects food production, quality and nutritional value, and safety. Soil nutrient loss is a major soil degradation process threatening nutrition and is recognized as being among the most important problems at a global level for food security and sustainability.

3. Soil physical degradation includes crusting, compaction, soil sealing and waterlogging, etc.
4. Soil biological degradation includes loss of humus, loss of soil biodiversity or lowering of soil fauna/flora (esp. lowering of the population, activity and diversity of soil micro-organisms).

1.5.1.2. Causes of Land Degradation

World soil health is under pressure from several threats including erosion, loss of soil organic carbon and biodiversity, pollution, and salinisation. Soil nutrient loss is a major soil degradation process threatening nutrition and is recognized as being among the most important problems at a global level for food security and sustainability.

The global pressures on agricultural lands are often anthropogenic in origin, for example, climate change, erosion, or land-use change (FAO, 2015; Smith et al., 2016). Pressures, for example, deforestation, overfishing and water scarcity are all working in the same direction. They are undermining and degrading nature.

- Deforestation and removal of natural vegetation and crop residues
- Overgrazing
- Inappropriate agricultural practices /exploitative use of the land/ such as:
 - Cultivation of marginal lands (steep slopes, shallow soils, etc)
 - Overcultivation (excessive tillage) or continuous cultivation
 - pesticides and fertilizer overuse
- (Bio-)Industrial activities and mining
- Urbanisation and (unsustainable) infrastructure

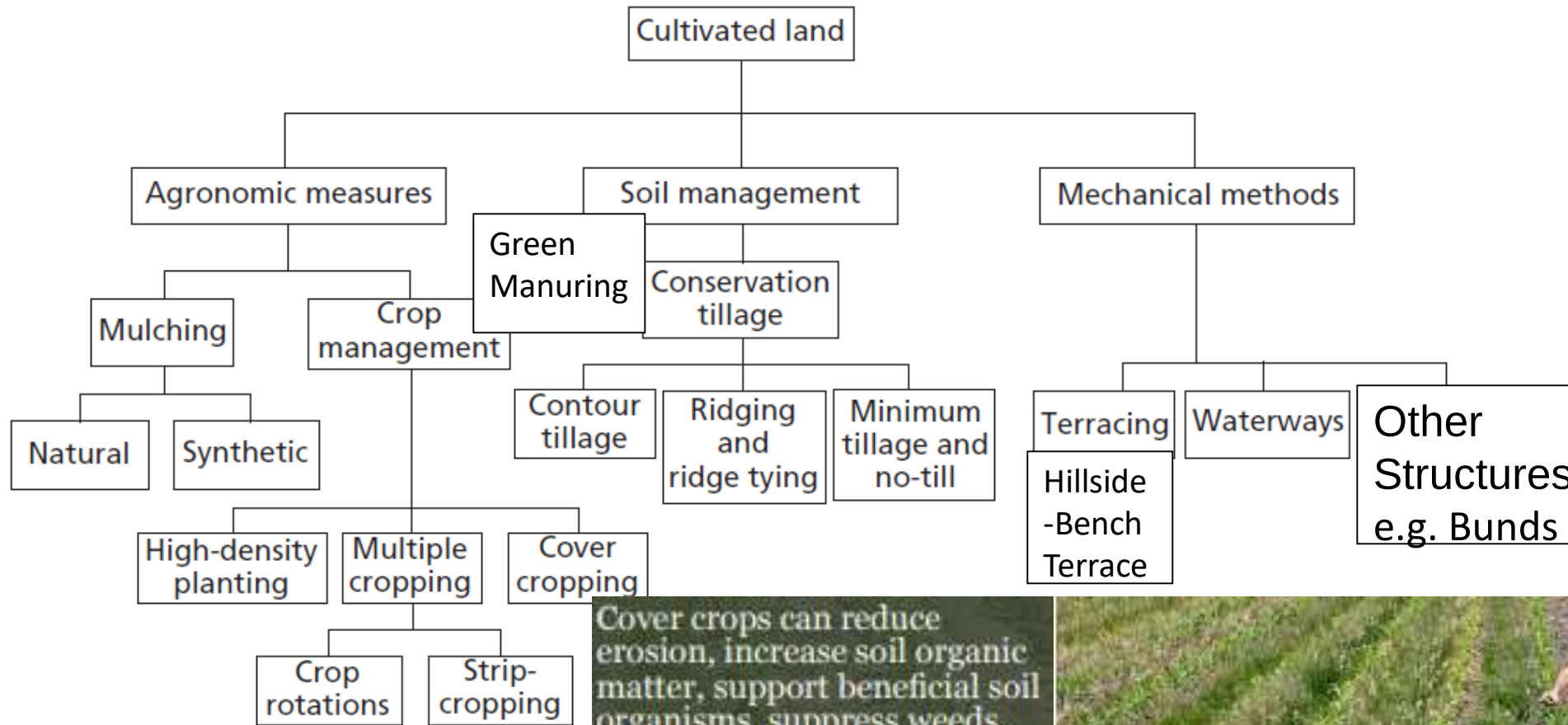
1.5.2. Environment/NRs Conservation

1.5.2.1. Soil and water conservation

- In hilly or mountainous areas, soil erosion is unavoidable due to rain and wind. Soil conservation is becoming increasingly important due to climate change and harmful farming practices.
- Soil and water conservation are those activities which maintain or enhance the productive capacity of the land including soil, water and vegetation in areas prone to degradation through
 - prevention or reduction of soil erosion,
 - conservation or drainage of water and
 - maintenance or improvement of soil fertility. Simply increasing soil organic matter alone improves moisture holding capacity and soil structure, so when it rains, more of it soaks into the ground, reducing flooding.
- Soil conservation practice includes soil management, crop management, engineering, range management and forestry operations.
- Terracing is one of the engineering practices used to prevent rainfall runoff in sloping lands for soil and water conservation.

...SOIL AND WATER CONSERVATION MEASURES

Fig 5. Soil conservation strategies for cultivated land



Cover crops can reduce erosion, increase soil organic matter, support beneficial soil organisms, suppress weeds, and release chemicals that are toxic to soilborne pathogens.

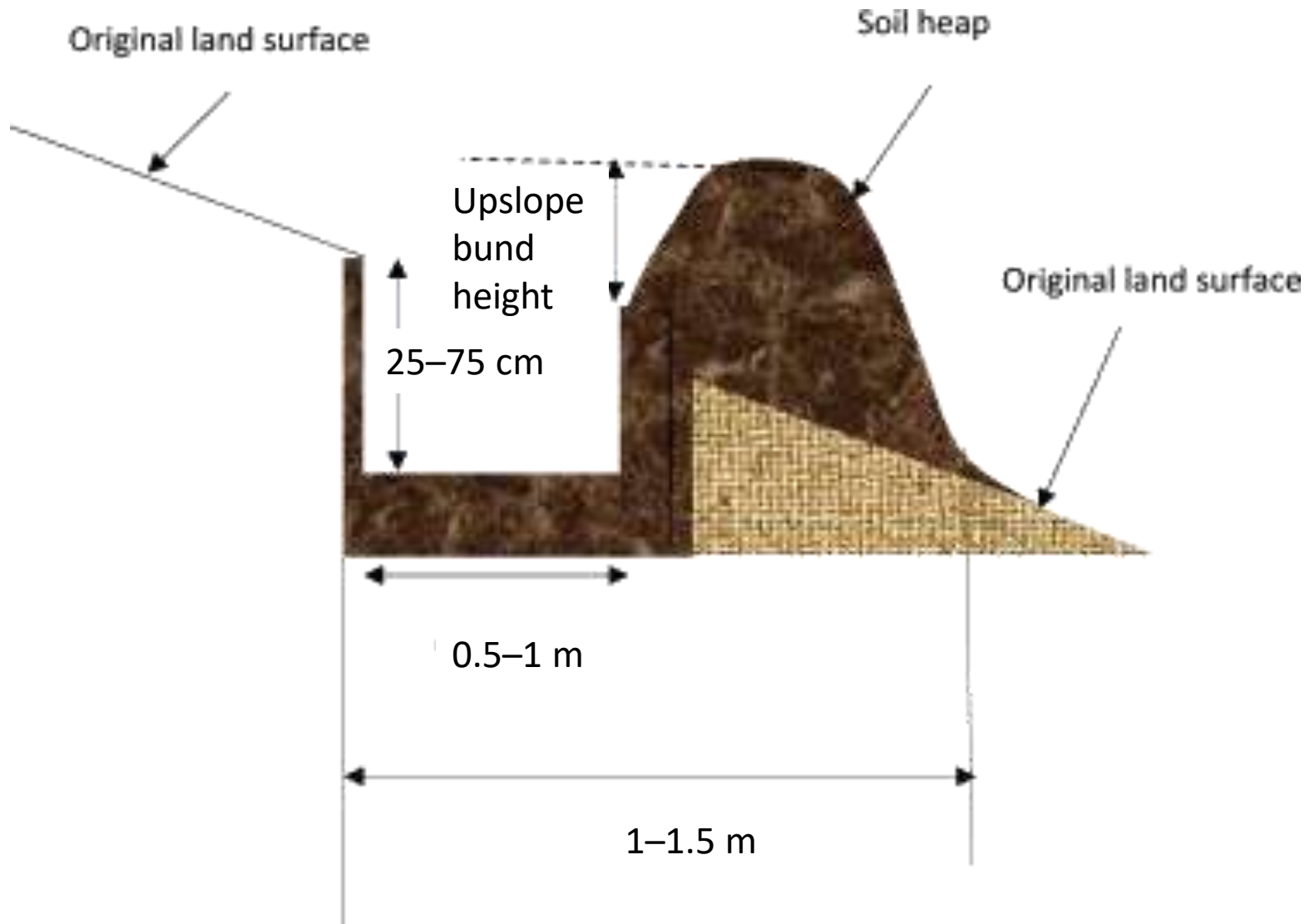
Cover crops also enhance soil biodiversity.

Multispecies cover cropping better manages soil health and improve sustainability.



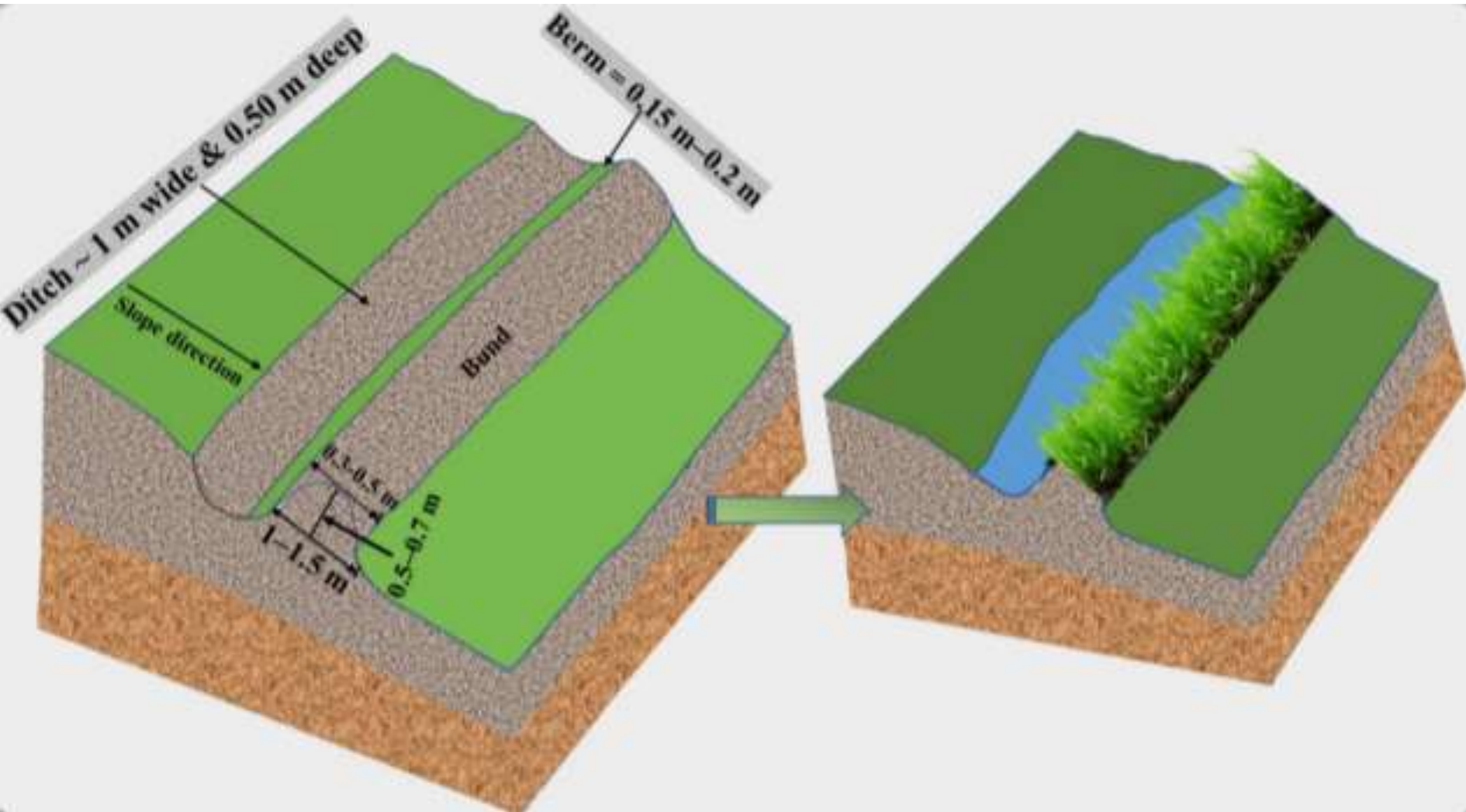
...SOIL AND WATER CONSERVATION MEASURES

Fig 6a. Side view and dimension of a typical soil bund



...SOIL AND WATER CONSERVATION MEASURES

Fig 6b. Side top view and dimension of a typical soil bund of soil bunds before (left) and after (right) being reinforced by grass or legumes



... 1.5.2.1. Soil and water conservation

2. CONSERVATION MEASURES ON GRASSLAND

- Controlled Grazing
- Zero-grazing: *Cut and Carry (stall-feeding)*
- Grassland Improvement



3. CONSERVATION MEASURES ON FORESTLAND

- Hillside Terrace
- Tree Planting

⋮

4. CONSERVATION MEASURES COMMON TO ALL LAND USE TYPES

- Area Closure
- Cutoff Drain
- Revegetation
- Water Harvesting
- Waterway
- Trash line

⋮

1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

1. Role of (Agro)forestry Practices for nutrition

- In association with food and agriculture, NWFPs/NTFPs provision includes
 - ✓ foods (nuts, fruits, mushrooms, honey, game, gums);
 - ✓ food additives (spices, herbs, flavorings, sweeteners); and
 - ✓ Fodder
- Agroforestry is also one of the best uses of agro-biodiversity that generates multiple benefits, i.e. ameliorates soil fertility, controls erosion and enhances moisture retention which
 - improves productivity and production of diversified food products and income for households and communities. E.g. Alley cropping is an agroforestry system in which food crops are grown in alleys between rows of hedges and it can be used as SWC practice on cultivated land.
- Inclusion of nutritionally rich/dense and diversified plants in the agroforestry system is essential to address malnutrition. This can also support the future food needs of the growing population through
 - ensuring people to have access to adequate and
 - nutritious food produced in an environmentally and socio-culturally sustainable manner.

... 1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

2. Link between soil management and nutrition

- Any thriving natural or agricultural ecosystem begins with soil. Thus, soil health is the starting point for the transformation of the entire food chain/web. Healthy and nutritious food for crops, livestock, and human starts with healthy soils.
- Improving soils can boost crop production by 40%. A balanced supply of soil nutrients is essential to support crop and forage grass yields and the nutritional value of the agri-food system. The lack of basic nutrients, i.e. macro and micronutrients, leads to the underdevelopment of plants and decrease in yields and crop nutritional value. Soil macro- and micronutrient deficiencies limit plant health, crop productivity and the nutritional quality of food, which together ultimately affect nutrition and human health.
- Soil fertility has a great role in producing sufficient, safe, and more nourishing food for healthier plants, animals, and people. Improved soil management yields a fertile (including high nutrient content) soil which in turn is mirrored in the crops/plants. Soil fertility and nutrition involve processes at scales ranging from molecules to the entire planet. Generally, soils offer a more nutritious agrifood system for enhancing human health and wellbeing while protecting the environment.

... 1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition ...

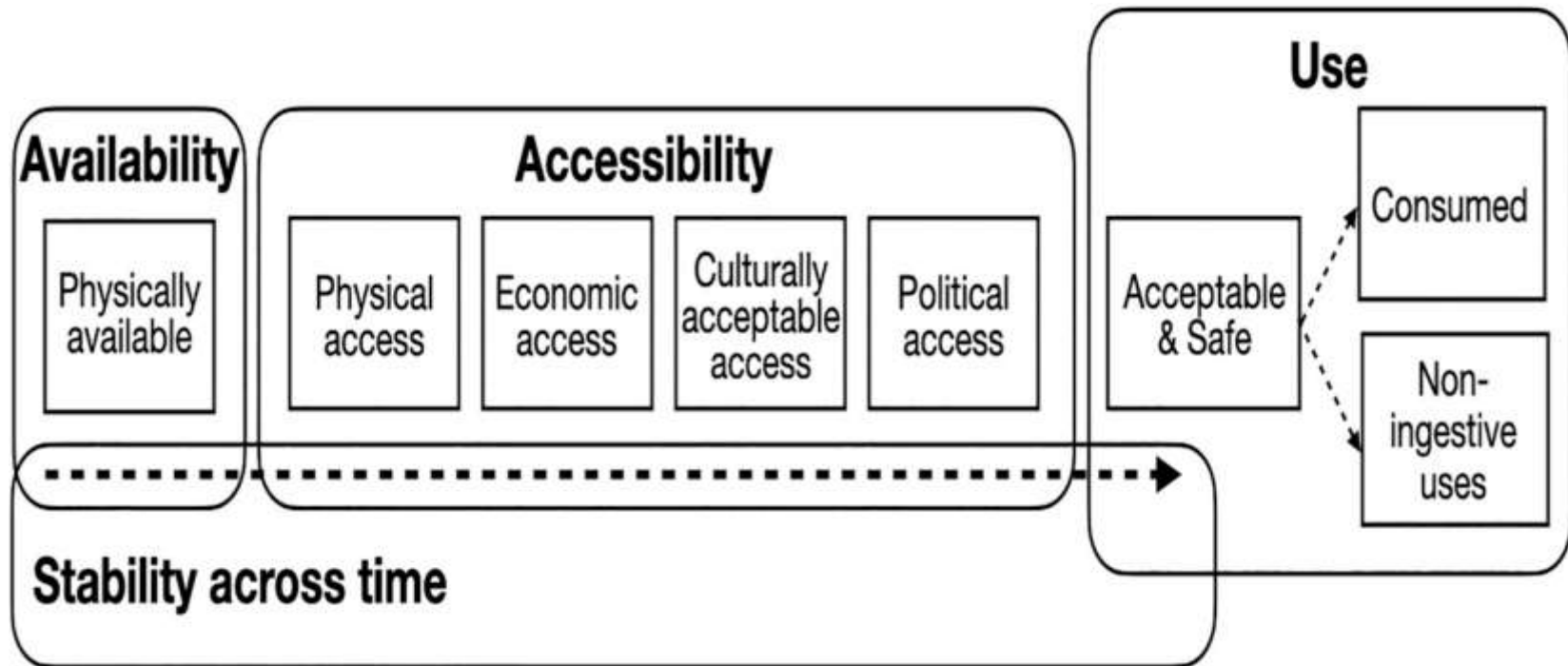
3. Irrigation and water conservation/harvesting for nutrition

- Irrigation increases food production and availability. Irrigation schemes are likely to provide enhanced food security and nutrition to farmers due to a greater availability and stability of food supplies and crop diversification.
- The use of irrigated agriculture enables crop production, often focused on high-value crops, during the dry season; specifically, it increases the number of harvests to 2-3 per year and leads to increased yields and crop diversification. This contributes to the improvement of nutrition.
- Smallholder irrigation systems are mostly used to grow vegetables in the dry season;
 - consequently, vegetable consumption among irrigation users and their communities usually increases.
 - Irrigation and water harvesting may also provide the water requirement of fruit seedlings all-year-round.
 - Vegetables and fruits are rich in micronutrients and provide important benefits, especially for children.
- Water scarcity is one of the most pressing challenges of our time. As the largest water user worldwide, the agriculture sector is uniquely positioned to drive the transition towards sustainable water management.

... 1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition ...

... 3. Water Security for Food Security, Good Nutrition, and Well-being

- The 4 domains of water security are of importance for humanity are its availability, access, stability, and quality.



- Consumption-related water use includes water that is drunk as well as eaten. Water is found in all foods, and some cooking techniques (e.g., boiling, steaming) substantially increase their water content. The noningestive uses of water are far more numerous. These include practices for sanitation and hygiene—for example, washing hands, dishes, and foods; bathing; laundry; and sewage disposal.

...1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

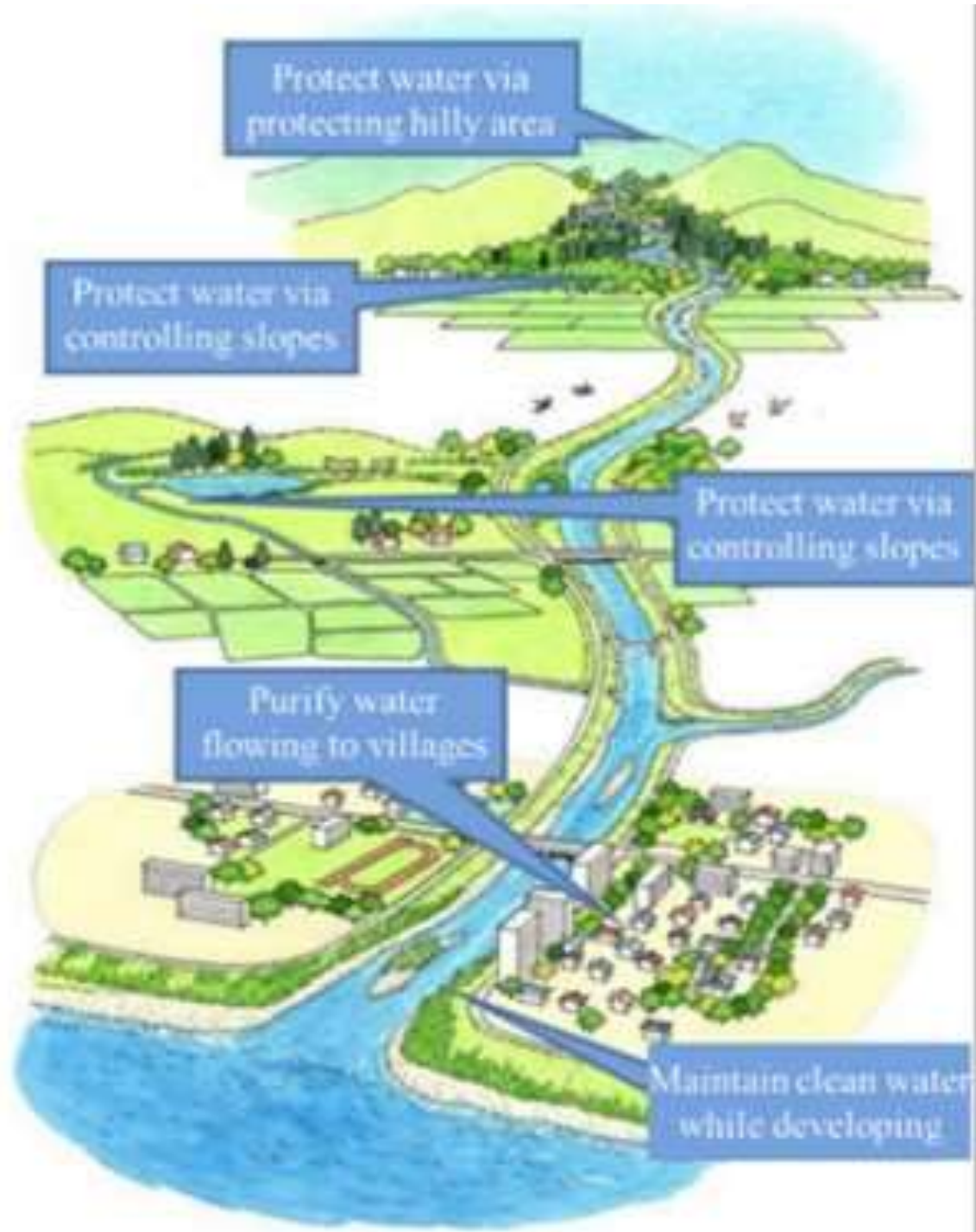
4. Watershed management and nutrition

- Watershed is an area of land that drains water to a common outlet.
- Through the creation of ecologically clean (eco-clean) watersheds, the soil and water loss control and economic development can be achieved in a coordinated way to better benefit the local people.
- The pattern/procedure for the creation of eco-clean watersheds is governing a clean small watershed, revitalizing a village, creating a demonstration project, cultivating a green industry, and making people rich.
- The eco-clean watershed management modes are **protecting water** via
 1. controlling hilly area - control pollution by reducing sources, intercept sewage and purify water – ecological restoration and
 2. protecting hilly area - protect water via controlling slope – purify water flowing to villages – maintain clean water quality while developing (Fig. 7).

...1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

...4. Watershed management and nutrition

Fig 8. Ecological clean (eco-clean) watershed management modes



...1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

...4. Watershed management and nutrition

- Sound management of watersheds provides multiple benefits in terms of food production, environmental health, and nutrition.
- Properly managed watersheds (rangelands, forests, etc) enhance ecosystem services that boost agricultural productivity, increase access to clean water, replenish aquifers, mitigate damage from storms and floods, and reduce negative environmental impacts from agriculture.
- Therefore, developing and properly managing watershed could contribute to the improved nutritional status of smallholder farmers through
 - enhancing agricultural production and diversification.

...1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition

Significance of NR conservation to food and nutrition

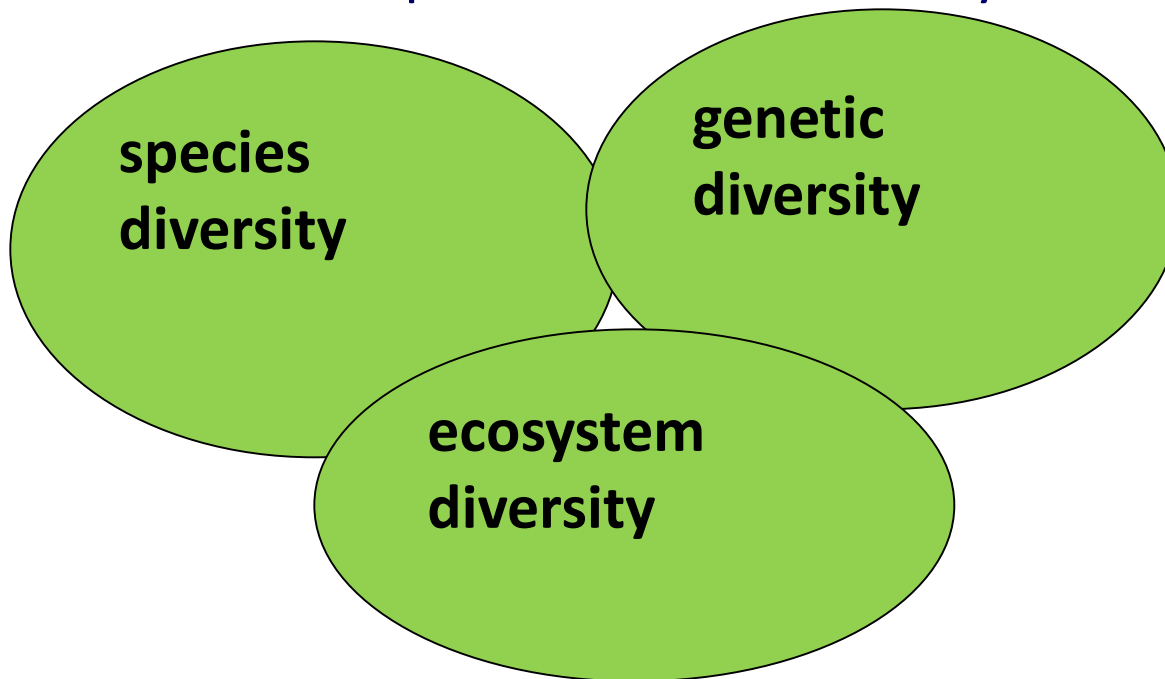
NR conservation e.g. By soil health, we can improve water quality and water quantity, and ultimately support a healthier planet. Moreover, the access and use of water affects many other nutrition-related phenomena, such as agricultural productivity, food-preparation practices, eating behaviors, dietary diversity, infant and young child feeding practices, and energy expenditure.

Significance of NR conservation to food and post-harvest

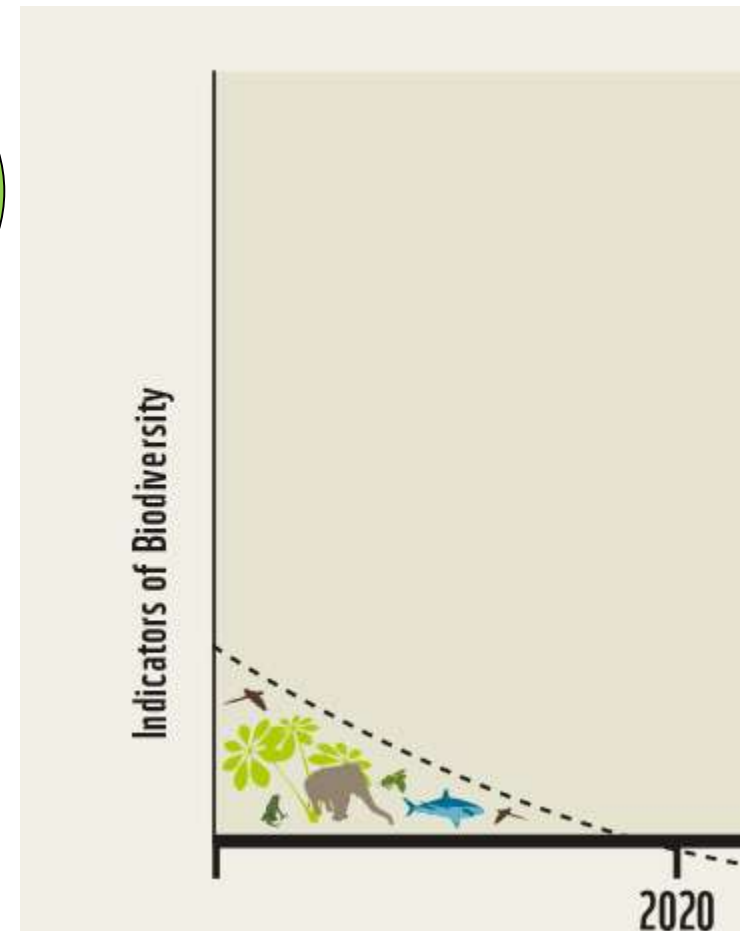
- The urgent challenge that the agricultural field is facing is to continue feeding the world's constantly growing population as resource availability steadily declines. Resources are used for producing food that will either never reach the consumer or will never get consumed.
- Farmers in a developing country lack the capital to invest in technologies, such as proper packing material and cooling systems.
- Natural resources conservation helps to decrease this food loss and post-harvest in two ways.
 1. By using resources more efficiently
 2. Recycling as much as possible OR turning waste into wealth/treasure

1.5.3. Biodiversity and Nutrition

- Agricultural biodiversity is important for food and nutritional security.
- The components of biodiversity are



A species range is an area where a particular species can be found during its lifetime. Species ranges include areas where individuals or communities may migrate or hibernate. Several factors determine species range; Climate is one important factor.



1.5.3.1. Importance and Loss of Biodiversity



Biodiversity gives
the **food we eat**,



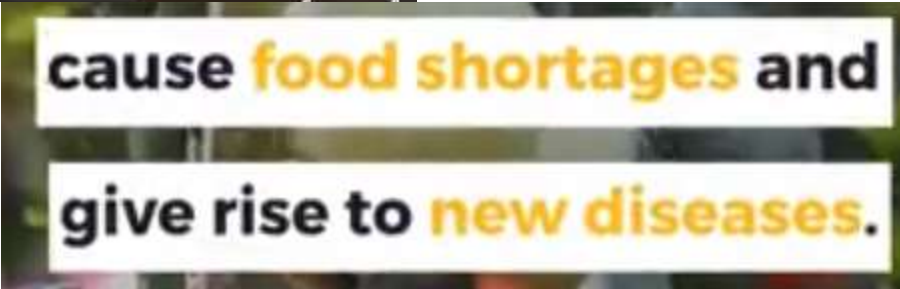
air to breathe and
water to drink.



80% of biodiversity on
land is in forests



The loss of biodiversity will
raise global temperatures,



cause **food shortages** and
give rise to new diseases.



Scientists say that around 27000
species of life forms are becoming
extinct every year due to loss of
habitat.



Loss of biodiversity further disrupts the
soil habitat and prevents soil
regeneration.

1.5.3.2. Biodiversity and Nature-positive Agriculture

- Agriculture takes up space. Prime farmland — land with good soil and water access — is a limited resource. These same areas often support rich wild ecosystems like prairies and forests; converting these areas into farms eliminates much of that wild biodiversity. Unfortunately, agriculture's continually expanding footprint places these sensitive and important wild areas at risk of destruction. This process of bringing more wild land into agriculture (to produce more food) is called extensification.
- Monocultures may offer consistency to farmers and the food industry, but they pose a risk to food security.
- Expansive agriculture is responsible for so much habitat loss, preventing wild lands from being converted into farmland is critical to maintaining biodiversity.
- Wild populations are a store of potentially useful genes that have been bred out of their cultivated relatives.
- Nature-positive development can only be achieved through transformative change, addressing the underlying societal factors that are the root causes of biodiversity loss and that drive development. A whole-of-society approach (to Biodiversity Governance) recognizes and builds on a society-wide mobilization of actors, ranging from non-state actors (such as indigenous peoples and local communities, NGOs, and the private sector) to all levels of government.

1.5.3.3. Biodiversity Conservation

- **Biodiversity** conservation can be beyond classical on-farm conservation approaches (e.g. conservation of local crop varieties and crop wild relatives) and may focus instead on the contributions of agroecology to mitigate species decline and ecosystem degradation at the landscape scale. This although important is less explicitly recognized within existing narratives of agroecology or conservation communities.
- By embracing both traditional knowledge and new research, farmers and scientists are producing food in a way that harnesses biodiversity to make the most of what nature provides. This approach is called agroecology, and is a core component of regenerative agriculture, which builds up natural resources like healthy soil and water rather than using them up.
- The adoption of safe farming methods and the use of farmer-owned seeds traditionally used in the country are the only ways to ensure food security (a continuous supply of food for the population). The **typical** diversity of seed in an household seed bank should include several grains (including farmer varieties) of maize, sorghum and pearl millet; legumes, which are key to food security and soil fertility; traditional fruit varieties; and nuts. Peanuts can be used in place of oil.

1.5.3.4. Importance of Biodiversity Conservation

- The biodiversity contributes to agriculture and food production in one way or another, is crucial to food security, nutrition and health, sustainable development and the supply of many vital ecosystem services. Biodiversity and healthy ecosystems are critical to food systems. Biodiversity includes a wide diversity of genetic resources which underpin vibrant food systems.
- Better conservation of biodiversity will have positive consequences on reducing malnutrition , and thus, poverty. Pollinators affect crop yields, thus market prices, and the nutritional value of foods. Pollinating animals like bees are so crucial for producing nutrient-rich crops that more than half of the population in some developing countries would experience malnutrition (nutrient deficiencies, especially of Vitamin A) if pollinators disappeared. The very diverse species that live in soils make up the often overlooked biodiversity that maintain healthy soils, helping micronutrients reach plants and ultimately our plates.
- ❑ The challenges of halting biodiversity loss and meeting future food requirements are extremely linked. So, halting biodiversity loss requires immediate action. Halting and reversing biodiversity loss requires a sharp increase in the area of natural ecosystems where biodiversity can thrive, through stronger conservation and restoration actions.
- Mainstreaming biodiversity in society and all economic sectors can sustainably address the issue of malnutrition. Embedding biodiversity conservation into land-use planning and food production systems is also an opportunity to achieve food and nutrition security (SDG 2) and meet the Paris Climate Agreement targets.

4. CLIMATE CHANGE IN AGRiculture and INTERVENTIONS

- **Weather refers to short term atmospheric conditions while climate is the weather of a specific region averaged over a long period of time.** Climate change refers to long-term changes and can alter weather patterns and climate.
- Agriculture is a risky business that is frequently exposed to uncertainties and is still the most vulnerable sector to climate change.
- Climate change increases the probability of extreme events and catastrophic risks affecting agriculture.
- Climate crisis is projected to increase global temperatures by 1-2°C, from the pre-industrial level, by 2050. For every 1°C increase, the global yields of the three major cereals— rice, wheat, and maize— are expected to decrease by 3-7%.
- Africa is suffering more than any region to the effects of extreme weather events and climate change. Over the past year (2022), Africa has experienced five of its top 30 deadliest weather disasters since record-keeping began in 1900.
- While the effects of climate variability are being felt on the field, the vast majority of African farmers have little or no capacity, to deal with climate change on their own, as the impact of climate change has destroyed the social and economic sustainability of family farming and has aggravated the food security challenges of a rapidly growing population.

4.1. Introduction to Climate Change

- is modification of earth's climate brought about as a result of changes in the atmosphere as well as interactions between the atmosphere and various other geologic, chemical, biological, and geographic factors within the earth system.
- is attributed to human activity, altering the composition of the global atmosphere. How we invest and consume will influence whether we meet the climate challenge and whether threatened communities and ecosystems can adapt and survive.
- Human activity includes the pollution that arises from industrial activity and other sources that produce greenhouse gases (GHGs).
- These gases, such as carbon dioxide, have the ability to absorb the spectrum of infrared light and contribute to the global warming of our atmosphere.
- Once produced, these gases can remain trapped in the atmosphere for tens or hundreds of years.
- Climate change is already happening, and is projected to continue in the future.

... 4.1. Introduction to Climate Change

- Human-induced climate change is causing dangerous and widespread disruption in nature and affecting the lives of billions of people around the world. It hits the most vulnerable people and ecosystems the hardest. E.g. it has exposed millions of people to acute food and water insecurity, especially in Africa. Africa tends to bear the burden of these crises and the harsh realities of climate change, even though it contributes little to their causes.
- The world faces unavoidable multiple climate hazards over the next two decades with global warming of 1.5°C (2.7°F). Even temporarily exceeding this warming level will result in additional severe impacts, some of which will be irreversible. Risks for society will increase, including infrastructure and low-lying coastal settlements to deep psychological impact.
- Every increment of warming results in rapidly escalating hazards. More intense heatwaves, heavier rainfall and other weather extremes e.g. **droughts and floods** further increase risks for human health and ecosystems exceeding plants' and animals' tolerance thresholds. In every region, people are dying from extreme heat. Climate-driven food and water insecurity is expected to increase with increased warming.

... 4.1. Introduction to Climate Change

Figure Climate change has already caused widespread impacts and related losses and damages on human systems and altered ecosystems worldwide.

Water availability and food production

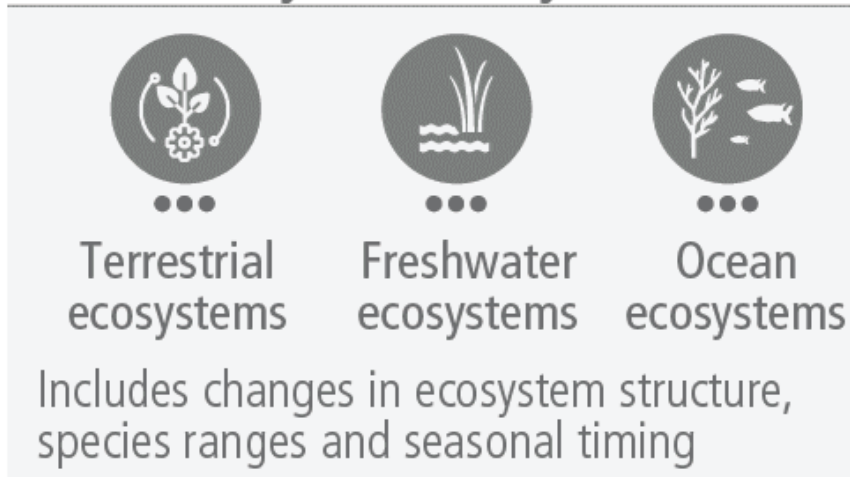


Health and well-being



Physical water availability includes balance of water available from various sources including -ground water, -water quality and -demand for water.

Biodiversity and ecosystems



Climate change influences cyclical timing patterns in 2 main ways

1. via overall warming, e.g., increases in mean annual temperature, which influences rates of biological processes such as development. E.g. flowering for plants and breeding or migration for animals.
2. via entire “climatic” growing seasons as potentially becoming more variable.

... 4.1. Introduction to Climate Change

Sources of emissions

- Biological and abiological processes in soil represent a major source of GHGs. Greenhouse gases (GHGs), mainly Carbon Dioxide (CO_2), Methane (CH_4) and Nitrous Oxide (N_2O), released from soils into the atmosphere are primarily biogenic by origin.
- Even if, deforestation due to expansion of pasturelands and croplands for livestock production contributes, the feed production stage contributes the greatest fraction (almost half) of total livestock (across the complete livestock production process) GHG emissions.
- *Ruminant livestock generates a significant proportion of anthropogenic GHG emissions*, and thereby contribute to climate change. Ruminants such as cattle, sheep, and goats undergo enteric fermentation – a digestive process in which microbes decompose and ferment food in the digestive tract or rumen. This process produces methane which is emitted via belching. Livestock accounts for a third of total global emissions of methane, a potent gas that traps heat about 80 times more effectively than carbon dioxide but breaks up more quickly in the atmosphere.
- The intensity of GHG emissions per dairy cow is roughly equal to those of two beef cattle or 16 pigs.

Food loss and waste (FLW) vs climate

- Food waste remains a significant challenge for many developing countries and SSA is significantly affected. Over one-third of the world's food is lost or wasted contributing 8 to 10 percent of global greenhouse gas (GHG) emissions. In low- and middle-income countries, over 40 percent of food loss occurs before a crop even makes it to market, whether due to inadequate storage in hot or humid climates, pests or microbes, spoilage, spillage in transport or otherwise.
- When food rots, it emits methane; thus, Food loss and waste (FLW) is linked to environmental, social and economic impacts.
- Fruits and vegetables are among the healthiest but most wasted foods. Given their perishable nature, great care, attention, and cooperation along all parts of fruit and vegetable value chains are required to reduce FLW. So, reducing losses and waste will not only contribute to our environment, but increase the availability of fruits and vegetables, and improve access to healthy diets
- Eliminating FLW would provide enough food to feed two billion people, as well as reduce GHG emissions.
- Despite increasing production, many still go hungry and adequate transportation is identified as one of the core barriers to reducing food waste and consequently hunger. A digital system approach can enhance communication between agro-food actors, including the digital decentralisation of transport procurement. The positive outcomes are shortened lead /process/ times, better asset utilization and reduced waste.

4.2. Climate Change and AGRiculture

- The global climate crisis is also a food security crisis, a water crisis, and a malnutrition crisis. In the last 60 years, climate change has reduced agricultural productivity growth by an average of 21%— and up to 40% in some regions— with impacts predicted to accelerate in coming decades.
- The climate and ecosystems are becoming increasingly disrupted. Climate and weather-related hazards, ecosystem damage, and loss of ecosystem services are occurring due to climate change and are expected to grow. Desertification is growing and an increasing population faces water stress, especially in Asia and Africa.
- Climate change will also weaken soil health and ecosystem services such as pollination, increase pressure from pests and diseases, and reduce marine animal species diversity, undermining food productivity in many regions.
- Since the 1960s, agricultural productivity has been reduced by at least 34% in Africa, directly due to climate change, more than any other region. Should warming be allowed to hit the 3°C mark – the current warming trajectory we are on – all present-day cropping areas for maize, millet and sorghum in Africa will become unsuitable.

4.2. Climate Change and AGRiculture

- There is vicious and reciprocal cycle between food systems and climate change with adverse impact on food and nutrition security.
- In the past 40 years, agricultural production has doubled, despite unsustainable food systems characterized by monoculture with excess fertilizer use, mass livestock production, and deforestation which increased GHG emissions and attendant climate change and loss of biodiversity. This in turn result in reduced food production. Climate change has impacted food systems through weather events such as drought, flooding, and heat waves with attendant loss of life, livelihood, and reduced productivity related to lower soil fertility, disrupted rain patterns, and acid rain from heavy fertilizer use. This vicious cycle leads to food insecurity and malnutrition in all its forms, environmental damage, water scarcity, and the emergence of new human, plant, and animal diseases.

4.2. Climate Change and AGRiculture

4.2.1. Effects of climate change on food system

- Climate change exacerbates the global food crises and vice versa, particularly in developing countries. Climate disasters have damaged food production in almost every region of the world. Climate change will affect every aspect of food production. The global food system both drives and is impacted by climate change. About a third of total GHG emissions are attributable to the food system including agriculture and land use. Land use contributes about one-quarter of GHG emissions, notably
 - CO₂ emissions from deforestation,
 - CH₄ emissions from rice and ruminant livestock and
 - N₂O emissions from fertiliser use.
- Climate change is already affecting food security through
 - increasing temperatures,
 - changing precipitation patterns, and
 - greater frequency of extreme events.

4.2. Climate Change and AGRiculture

...4.2.1. Effects of climate change on food system

- The climate emergency affects us all deeply, in so many ways; it undermines all aspects of the food systems more drastically.
- Global warming is the most influential long-term trend affecting the agriculture sector. Climate change is putting an unprecedented pressure on natural ecosystems, affecting water availability, food production, and biomass sourcing.
- Many consequences of climate change threaten food security and diet quality, and exposing vulnerable populations to multiple forms of malnutrition.
- Climate change worsens unsustainable food systems by directly impacting soil fertility, rain patterns/water availability, reduced food yields and food production, nutrient concentration/composition, and bioavailability of nutrients in foods, and increased pests and diseases.
- Further problems arise from indirect impacts such as pests that result in increased occurrence of spoilage and food safety hazards at various stages of the food chain from primary production to post-harvest protection through to consumption. Each of these factors may have deleterious impacts on human nutrition.
- Climate change amplifies the impacts of other risks, such as extreme weather and pests and diseases. Intertwined with this are negative impacts from biodiversity loss and landscape degradation.

4.2. Climate Change and AGRiculture

4.2.2. Impacts of climate on food and nutrition security

- Climate change presents an unprecedented threat to nutrition and food security globally, and food systems in their current state are a major driver of the crisis.
- Extreme weather devastates crops and livestock, and destabilises the very water cycle that they are intimately a part of.
- Increased temperature, drought, rainfall variability, extreme weather and ocean acidification associated with climate change is related to reduced food production, altered nutrient content, inequitable access to healthy foods and beverages, and high rates of food insecurity. Climate change poses a threat to both food security for communities as well as nutrition security.
- Nutrition security builds on food security by connecting how the quality of what we eat can help reduce diet-related diseases. It also emphasizes equity and tackling long-standing health disparities. Climate change and its varied effects, including extreme weather events and longer-term degradation and loss of resources, can create new insecurities and exacerbate existing inequities.

4.2. Climate Change and AGRiculture

...4.2.2. Impacts of climate on food and nutrition security

Climate change further imposes threats to food systems by impacting the quantity, quality and affordability of food. E.g. there are differences in the distributions of diet diversification /diversity/with changing rainfall amount and rainfall variability.

If global temperatures rise beyond 1.5°C, this would lead to a more severe risk to food security, increasing ten-fold from 8 million to 80 million people facing severe hunger between low-emissions and high-emissions scenarios by 2050.

- Food Systems: Local, regional, and global food systems are weakened because of emerging climate change impacts on agricultural productivity. At the local level, climate change threatens Indigenous food systems and the production of and access to traditional food.
- Food Safety: Food safety and processing is impacted by availability of water and extreme weather events. Water stress will impact water costs and, therefore, the costs of properly cleaning food products. Extreme weather can cause disruptions to food processing and distribution and may expose food to unsafe temperatures. Such disruptions can directly affect food preservation and safety, which has serious consequences for health.

4.2. Climate Change and AGRiculture

...4.2.2. Impacts of climate on food and nutrition security

- **Food Access:** Transportation, storage, preservation, and access points (markets, grocery stores, food banks) to nutritious, safe food will be impacted both by rising temperatures and severe weather events that interrupt supply chain infrastructure. Limited food access has direct implications for community health, particularly within most vulnerable communities.
- **Food Loss and Waste:** Food loss during processing and distribution is likely to increase because of rising temperatures, leading to increased waste as food moves from market to consumer. If food reaching consumers is not at peak freshness, food waste at the consumer level will likely rise. An increase in food waste has effects on food access, community health, and agroecosystem sustainability.
- **Nutritional Quality:** The impacts of climate change on the nutritive quality of food are not yet fully established; however, several correlations, some nonlinear, have been observed. Food is threatened by climate change impacts on soil, water quality and availability, and rising temperatures, all of which may have complex effects on the nutritive quality of a given product.
- **Most Vulnerable Communities:** Food and nutrition access and safety are particularly insecure in and among most vulnerable communities. The impacts of climate change will exacerbate existing inequities. The cascading effects of the loss of assets, morbidity, and displacement caused by climate change can further compound access to safe, nutritious, and culturally appropriate food for these communities.

4.2. Climate Change and AGRiculture

...4.2.1. Effects of climate change on food system

- Climate change has increased risks to global food security— slowing down rates of productivity, and even reducing nutritive content. Climate change worsens under-nutrition by affecting all stages of the food value chain.
- In many regions of the world, particularly in Africa south of the Sahara and in South Asia,
 - agriculture will have greater difficulty in producing enough food that is of good quality (diverse, rich in nutrients, and safe).
- Future warming will negatively affect food systems in Africa by shortening growing seasons and increasing water stress -- this is already happening in Eastern Africa -- leading to famine. Food systems (and ability to move food from production to markets) will be affected, which will limit
 - consumers' access to diverse and high quality diets in their local food environments.

Without action, climate change will impact nutrition through decreased food quantity and access, decreased dietary diversity, and decreased food nutritional content.

4.2. Climate Change and AGRiculture

4.2.3. Effects of agriculture in climate change

“The food system based on the industrial agriculture narrative has failed to feed the world and Africa by generating toxic foods, polluting the environment, and fueling climate crises.”

Dr. Chris Macoloo, AFSA (Alliance for Food Sovereignty in Africa) Chair

- Rather than nourishing global health and wellbeing, **food systems** remain at the heart of the social and environmental **crises** we face. In pursuing our **intensive production of food**, we may be **threatening** our **health** and that of many other **species** on the planet, harming the environment and worsening the already difficult global scenario of **environmental degradation**, poverty, malnutrition, migration, disease, and **climate change**.

E.g. the destruction of natural ecosystems to meet human demand has led to large emissions: it is estimated that deforestation and peatland drainage contribute to about 15% of total global **greenhouse gas** emissions, while also reducing the area where carbon can be naturally sequestered in the future.

4.3. Climate-resilient interventions...

The current track will bring further intensification of extreme weather, ecosystem degradation and damage to lives and livelihoods.

The heat must be turned down; communities must adapt to those impacts of climate change that are already here.

A portfolio approach, investing in a wide range of promising solutions and minimizing risk, should be taken .

Taking an inclusive approach to carbon removals spurs private sector investment and leads to the scaling up of multiple solutions. Investing in areas (such as green urban infrastructure; halting deforestation; ecosystem restoration; sustainable food systems, including reduced food loss and waste) and more, will help to stabilize climate. Reduce nature and biodiversity loss, and pollution and waste – the other two prongs of the triple planetary crisis. Deliver many other benefits: from cleaner air and healthier nature to decent jobs and more equity.

We need a climate resilient development, a paradigm shift, with:

- **adaptation** - managing the risk that comes with a changing climate and increasingly common extreme weather events, and
- **mitigation** - reducing or avoiding GHG emissions.

...4.3. Climate-resilient interventions

Adaptation options

Mitigation options

LAND, WATER, FOOD	Env'l and NR manage't	Adaptation options		Mitigation options	
		Efficient livestock systems	1	Reduce conversion of natural ecosystems	
		Improved cropland management	2	Carbon sequestration in agriculture	
		Water resource management	3	Ecosystem restoration, afforestation, reforestation	
		Biodiversity management and ecosystem connectivity	4	Shift to sustainable healthy diets	
		Agroforestry	5	Improved sustainable forest management	
		Sustainable aquaculture and fisheries	6	Reduce methane and N ₂ O in agriculture	
HEALTH		Forest-based adaptation	7	Reduce food loss and food waste	
		All are adapting production to climate change.		The most notable mitigation measures related to food systems is improving forest, cropland, and grassland management. Reducing tropical deforestation has the greatest potential for reducing GHG emissions of any single solution from any sector.	
HEALTH		Enhanced health services (e.g. WASH, nutrition and diets)		Indigenous peoples and other traditional knowledge holders have thrived living sustainably with limited resources, in some cases for thousands of years.	
		WASH - water, sanitation and hygiene			

Agriculture is the main sector affecting soil and landscape management, so improving agricultural management and policy that supports it is essential to reducing GHG emissions. This includes sequestering carbon in the soil.

Fig. Climate Actions in 1. Land, Water and Food and 2. Health

...4.3. Climate-resilient interventions...

Food, land, and water systems are at the forefront of the climate crisis, and transformative changes are urgently needed to reduce emissions, increase resilience and ensure food security for all.

But while food, land, and water systems are the nexus of climate change impacts, they are also a nexus of solutions, both in mitigation – reducing emissions, and adaptation – responding and building resilience and reducing inequity in a world facing increasing climate challenges.

While climate change has reduced food security, there are a number of adaptation options that can be effective in reducing its impacts. Those include more climate resilient crops, better water management and storage, irrigation, and farm and landscape level diversification; moreover, examples of food system-based adaptation measures such as cultivar improvements, on-farm water management and storage, soil moisture conservation, irrigation, agroforestry and agroecology.

When you understand agroecology in its fullness, it gives a pathway to transition to food and farming systems that are just, sustainable, climate resilient and regenerative, and economically viable.

The most cost-effective and sustainable options to cut emissions in the short and long term are halting deforestation and embarking on restoration of degraded nature as well as shifting to more plant-based diets.

4.3.1. Climate Adaptation in Land, Water and Food

Sustainable farming approaches can be regenerative agriculture, conservation agriculture, sustainable intensification, agroecology, and organic agriculture.

Regenerative agriculture is to mean farming that improves, rather than degrades, landscape and ecological processes such as water, nutrient and carbon cycles.

Agroecology combines the terms 'agronomy (agricultural science)' and 'ecology'.

Much more than an agricultural model, permaculture is presented as a philosophy of life that consists of being inspired by nature to finally reproduce what it does.

They make it possible to obtain a natural ecosystem where plants, insects and all microorganisms live in perfect symbiosis. Every living being is a vital link in the balance of the ecosystem.

To grow and eat healthy, you have to let the magic of nature work. This is the very principle of permaculture. For this, place each plant thoughtfully with the objective of allowing species to interact with each other to form a virtuous circle. For example, nettles can be placed in order to attract ladybugs which, in turn, will prevent aphids from attacking tomatoes and so on. Planting legumes will act as natural nitrogen fertilizers which will help other plants to grow. making your own natural fertilizers such as nettle manure, an effective repellent against aphids.

...4.3.1. Climate Adaptation in Land, Water and Food

They can include practices such as:

- integrating livestock into cropping systems to replenish soil and reduce the cost of animal feed and fertiliser as well as diversification of plant species
- conservation tillage - minimum soil disturbance and coverage with plants to retain carbon, moisture and nutrients and reduce erosion
- using less synthetic chemicals in farming.

Some examples of nature-based solutions (NBS) related to agriculture and grasslands include avoided grassland conversion, biochar, cropland nutrient management, conservation agriculture, trees in croplands, improved rice cultivation and different grazing practices related to animal management, optimal intensity, legumes in pastures and improved feed.

Potentially effective interventions may be improving management of rangelands and croplands to make the feed supply more sustainable; providing both early-warning alerts and adaptive safety nets for herders and livestock farmers ; and adding trees to rangelands in “silvopastoral” systems, where the trees provide shade to reduce heat stresses. It may also be possible to boost local incomes through carbon credit trading and benefit sharing when rangelands are restored.

An adaptation such as the modification of production and management systems involves

- diversification of livestock animals and crops (as in conservation agriculture),
- integration of livestock systems with forestry and crop production, and
- changing the timing and locations of farm operations.

...4.3.1. Climate Adaptation in Land, Water and Food

4.3.1.1. Efficient livestock systems

The livestock sector must contribute to addressing a range of challenges related to the environment (e.g. deforestation, land-use changes, greenhouse-gas emissions, unsustainable water and land use, pollution, food–feed competition), herd management (e.g. low productivity, overgrazing, poor animal welfare), animal health related issues (e.g. diseases, antimicrobial resistance), human-livestock related issues (e.g. zoonotic and food-borne diseases) and social issues (e.g. equity).

Management practices of cattle farming systems must be improved, particularly to increase the systems' feed self-sufficiency, food production and environmental performances.

Farm management practices vary widely around the world, and hence in regions with greater emissions per unit of animal protein produced there are substantial mitigation opportunities through the adoption of best practices. The abatement potential can be increased if the livestock production chain worldwide were to match the practices of the most efficient producers.

Changes in management may include improving herd health, and breeding for improved productivity. In the case of animal feed, options include processing feed grain to improve digestibility and feed supplements such as seaweeds *Asparagopsis taxiformis*, potentially in combination with *Oedogonium* sp. Improve animal health and husbandry through: feed changes and supplements; selective breeding to improve productivity and animal health/fertility.

By far the largest share of this abatement would come from cattle, i.e. from meat and from dairy.

...4.3.1. Climate Adaptation in Land, Water and Food

... 4.3.1.1. Efficient livestock systems

Improving feeding practices as an adaptation measure could indirectly improve the efficiency of livestock production. Suggested feeding practices include modification of diets composition, changing feeding time and/or frequency, incorporating agroforestry species in the animal diet, training producers in production and conservation of feed for different agro-ecological zones, new sources of protein for feed (e.g algae, insects, etc.) and feed additives - such as seaweeds - to suppress methane production. These practices can reduce the risk from climate change by promoting higher intake or compensating low feed consumption, reducing excessive heat load, decreasing the feed insecurity during dry seasons, and reducing animal malnutrition and mortality, respectively.

Shifting locations of livestock and crop production could reduce soil erosion and improve moisture and nutrient retention. Another adaptive measure could be adjusting crop rotations and changing timing of management operations (e.g. grazing, planting, spraying, irrigating). This measure can be adapted to changes in duration of growing seasons, heat waves and precipitation variability.

Regenerative livestock systems can solve environmental problems like climate change and soil degradation.

An improved variety of grass can increase soil carbon storage while reducing nitrous oxide emissions to offset the climate impact of livestock. Some grass species can also grow on marginal land, meaning farmers can make use of degraded soils to maintain livestock production even where food crops cannot grow.

Pay special attention to low-carbon technologies that can also improve milk production.

Improve productivity of small- and medium-sized dairy farms.

...4.3.1. Climate Adaptation in Land, Water and Food

4.3.1.2. Improved Cropland Management

Improved cropland management is mainly regenerative agriculture. Nature-friendly farming like organic and agroecology is essential. Farms where nature thrives in well managed habitats – healthy soils, thicker hedgerows, abundant trees, and diverse grasslands – are in a better place to maintain food production in the face of climate change. Agriculture must not be a missed opportunity to clearly address the nature emergency and support farmers to make the big changes that are urgently needed.

Agroecology requires constant learning, experimentation and adaptation resulting in huge rewards. E.g. recognition of the role of the farmer and the peer to peer learning model to drive increased regenerative practice. These can not only deliver the climate adaptation needed, but also wider global examples where they increasingly act as early warning systems for risks such as salinization of soils. Agroecology builds resilience and transforms communities in the face of intersecting crises, and offers a roadmap for a more sustainable and equitable future.

Testing climate-resilient varieties of staple crops that can both thrive in extreme conditions and reduce agricultural emissions.

From improved rice cultivation (e.g. improved water management or alternate flooding/drainage wetland rice), an NBS:

System of Rice Intensification (SRI) is a climate-smart agricultural methodology. It is a sustainable agricultural practice that provides ecological, social, and economic benefits to farmers, families, consumers, and businesses.

SRI follows four key principles that produce significantly greater crop yields, simultaneously minimizing inputs and GHGs, particularly methane. The principles are: the early establishment of healthy plants; low plant density; soil enrichment; and the sparing application of water.

...4.3.1. Climate Adaptation in Land, Water and Food

4.3.1.3. Environmental and natural resource management

The climate conversation is evolving as biodiversity and natural capital rise up the agenda. Producing more food not only isn't the answer to the food insecurity we face, but will only make the situation worse if we fail to tackle the climate and nature crises. Environmental and natural resource management is among the essential aspects and features of CSA. CSA strategies should protect natural resources to enhance agroecosystem resilience and natural biological processes, reduce soil erosion, and provide other ecosystem services. Adaptation strategies should reduce environmental pollution, conserve natural resources, such as water, soil, and biodiversity, and increase the yield and income of smallholder farmer.

Multiple aspects of diversity need to be nurtured and targeted at all scales. This includes the diversity of crops, agricultural practices and landscapes in food production systems.

Adaptation measures for agriculture systems include agroforestry, community-based adaptation, farm and landscape diversification, and urban agriculture.

There are thousands of indigenous and traditional African crops with untapped potential for climate adaptation and nutrition. In biodiversity management, use of Africa's forgotten food crops in the diversification of food supply for healthy diets in a changing climate is important.

Forest based adaptation includes sustainable forest management, forest conservation and restoration, reforestation and afforestation.

...4.3.1. Climate Adaptation in Land, Water and Food

...4.3.1.3. Environmental and natural resource management

Reforestation, afforestation and agroforestry enhance agroecological succession, maintaining the functionality and sustainability of farming systems. Additionally, forests play important roles in sequestering carbon, maintaining ecosystem services, and improving social adaptation to climate change impacts.

Use different processes for seedlings and to save water, including mulching which involves covering the soil by crop residue to maintain soil moisture.

Farmer Managed Natural Regeneration (FMNR) is a low-cost land restoration technique used to combat poverty and hunger amongst poor subsistence farmers by increasing food and timber production and resilience to climate extremes. In practice, FMNR involves the systematic regrowth and management of trees and shrubs from felled tree stumps, sprouting root systems or seeds.

The regrown trees and shrubs – integrated into crops and grazing pastures – help restore soil structure and fertility, inhibit erosion and soil moisture evaporation, rehabilitate springs and the water table, and increase biodiversity.

Products like Moringa have a market potential and are good for the health of humans and the planet.

4.3.2. Climate Change Mitigation

Large-scale land conversion for bioenergy, biochar, or afforestation can increase risks to biodiversity, water and food security. In contrast, restoring natural forests and drained peatlands, and improving sustainability of managed forests enhances the resilience of carbon stocks and sinks and reduces ecosystem vulnerability to climate change. Cooperation, and inclusive decision making, with local communities and Indigenous Peoples, as well as recognition of inherent rights of Indigenous Peoples, is integral to successful adaptation across forests and other ecosystems.

Natural rivers, wetlands and upstream forests reduce flood risk in most circumstances. Enhancing natural water retention such as by restoring wetlands and rivers, land use planning such as no build zones or upstream forest management, can further reduce flood risk. For inland flooding, combinations of non-structural measures like early warning systems and structural measures like levees have reduced loss of lives, but hard defences against flooding or sea level rise can also be maladaptive.

...4.3.2. Climate Change Mitigation

4.3.1.1. Carbon sequestration in agriculture

Climate-smart farming practices are quickly becoming a leading global strategy to adapt to and mitigate climate change (CC). Precision nitrogen and weed management, cover crops, reduced tillage and preventing burning of agricultural crop residues are the top climate-smart strategies to lower GHGs and increase carbon storage.

Agrifood systems offer a unique opportunity to address CC by:

1. building resilience across agrifood systems, we ensure their adaptation to CC.
2. reducing GHG emissions, if agricultural systems are managed sustainably.

Soils can hold vast quantities of carbon — greater than the atmosphere and vegetation combined — which helps the environment, yet the world's cultivated soils have lost more than 50-70% of their carbon stocks.

Healthy soils are needed to sequester carbon and thus reduce net greenhouse gas emissions, and for helping agriculture adapt to climate change.

Improved soil health benefits growers, the environment, and society by improving nutrient use efficiency, drought resilience, downstream water quality, greenhouse gas mitigation, carbon sequestration, and ultimately, grower profitability and quality of life.

Without investment in soil fertility, soil quality and in turn agricultural productivity will continue to deteriorate with climate change. We need both climate-adapted, nutritious crops and healthy soils for a food secure future.

...4.3.2. Climate Change Mitigation

4.3.1.2. Ecosystem restoration, reforestation, and afforestation

Conservation, improved management, and restoration of forests and other ecosystems offer the largest share of economic mitigation potential, with reduced deforestation in tropical regions having the highest total mitigation potential. Ecosystem restoration, reforestation, and afforestation can lead to trade-offs due to competing demands on land. Minimizing trade-offs required integrated approaches to meet multiple objectives including food security. Demand-side measures (shifting to sustainable healthy diets and reducing food loss/waste) and sustainable agricultural intensification can reduce ecosystem conversion and CH₄ and N₂O emissions, and free up land for reforestation and ecosystem restoration. Sustainably sourced agriculture and forest products, including long-lived wood products, can be used instead of more GHG-intensive products in other sectors.

Some options, such as conservation of high-carbon ecosystems (e.g., peatlands, wetlands, rangelands, mangroves and forests), have immediate impacts while others, such as restoration of high-carbon ecosystems, reclamation of degraded soils or afforestation, take decades to deliver measurable results.

...4.3.2. Climate Change Mitigation

...4.3.1.2. Ecosystem restoration, reforestation, and afforestation

One of the most impactful transformations to significantly reduce CO₂ emissions from global agriculture while preserving biodiversity and enhancing farmers' income is reclaiming degraded farmland to avoid land expansion.

Protection and restoration of coastal 'blue carbon' ecosystems (e.g., tidal marshes, seagrass meadows and mangroves) could reduce emissions and/or increase carbon uptake and storage. Coastal wetlands protect against coastal erosion and flooding. Strengthening precautionary approaches, such as rebuilding overexploited or depleted fisheries, and responsiveness of existing fisheries management strategies reduces negative climate change impacts on fisheries, with benefits for regional economies and livelihoods. Ecosystem-based management in fisheries and aquaculture supports food security, biodiversity, human health and well-being.

...4.3.2. Climate Change Mitigation

...4.3.2.2. Ecosystem restoration, reforestation, and afforestation

(almost) all land cover types can be targeted for planting trees:

- forests can be restored and reforested
- burnt and abandoned agricultural land can be afforested
- perennial crops, trees planted in combination with annual crops or pasture, and/or green infrastructure can spread across agricultural land
- In the urban environment (cities), green spaces – such as urban parks and urban orchards/forests, green buildings (specially, green walls and roofs), urban wetlands, community allotments, tree-lined avenues, **indoor farming systems and vertical farming** – can help render cities more sustainable, livable and resilient. Creating new public green spaces and upgrading existing ones should be a priority for a city and a key element of its climate action plan.

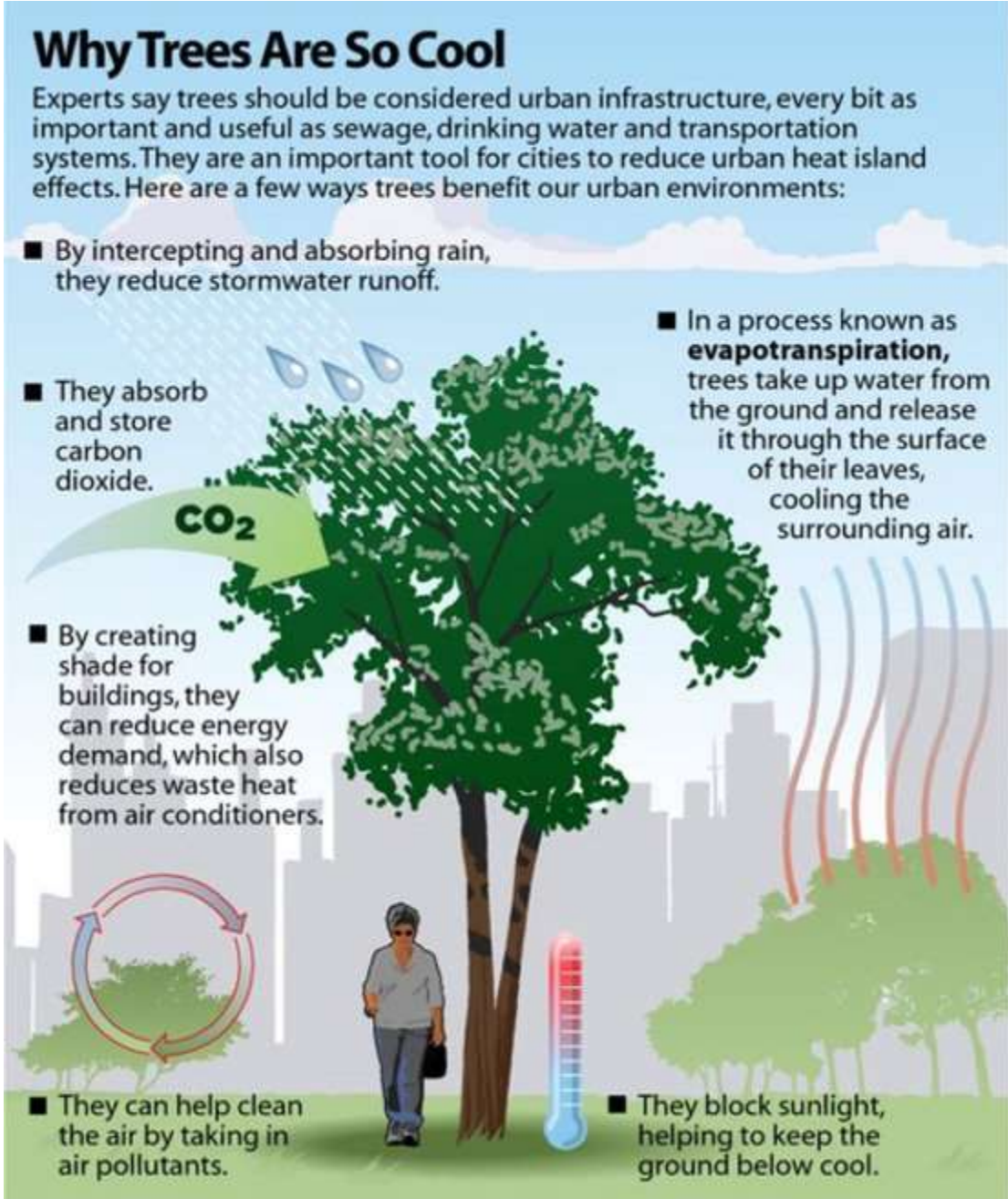
Moreover, urban gardens/greening

- has multiple benefits, such as biodiversity conservation, preservation of cultural values, and temperature regulation and role in developing ecosystem services.
- can also offer solutions to major urban environmental problems, such as air and water pollution, heat waves (and floods), and provide ample opportunities for recreation, relaxation, social interaction, and education.
- awaken admiration for nature, inspiration, and peace being intermediary between humans and nature; it restores harmony between humans and nature. So, nature will be 'up close and personal' rather than distant and abstract in the city.

...4.3.2.2. Ecosystem restoration, reforestation, and afforestation

Why Trees Are So Cool

Experts say trees should be considered urban infrastructure, every bit as important and useful as sewage, drinking water and transportation systems. They are an important tool for cities to reduce urban heat island effects. Here are a few ways trees benefit our urban environments:

- By intercepting and absorbing rain, they reduce stormwater runoff.
 - They absorb and store carbon dioxide.
 - In a process known as **evapotranspiration**, trees take up water from the ground and release it through the surface of their leaves, cooling the surrounding air.
 - By creating shade for buildings, they can reduce energy demand, which also reduces waste heat from air conditioners.
 - They can help clean the air by taking in air pollutants.
 - They block sunlight, helping to keep the ground below cool.
- 

4.3.2.3. Shifting to sustainable healthy diets or Dietary Shift

Shaping a resilient, healthy, and food-secure food system

The shift focuses on reduced meat intake, promoting a protein-rich diet with less saturated fat and cholesterol...

High Yields

2. Low Food Waste

Each year one third of global food production is lost or wasted.

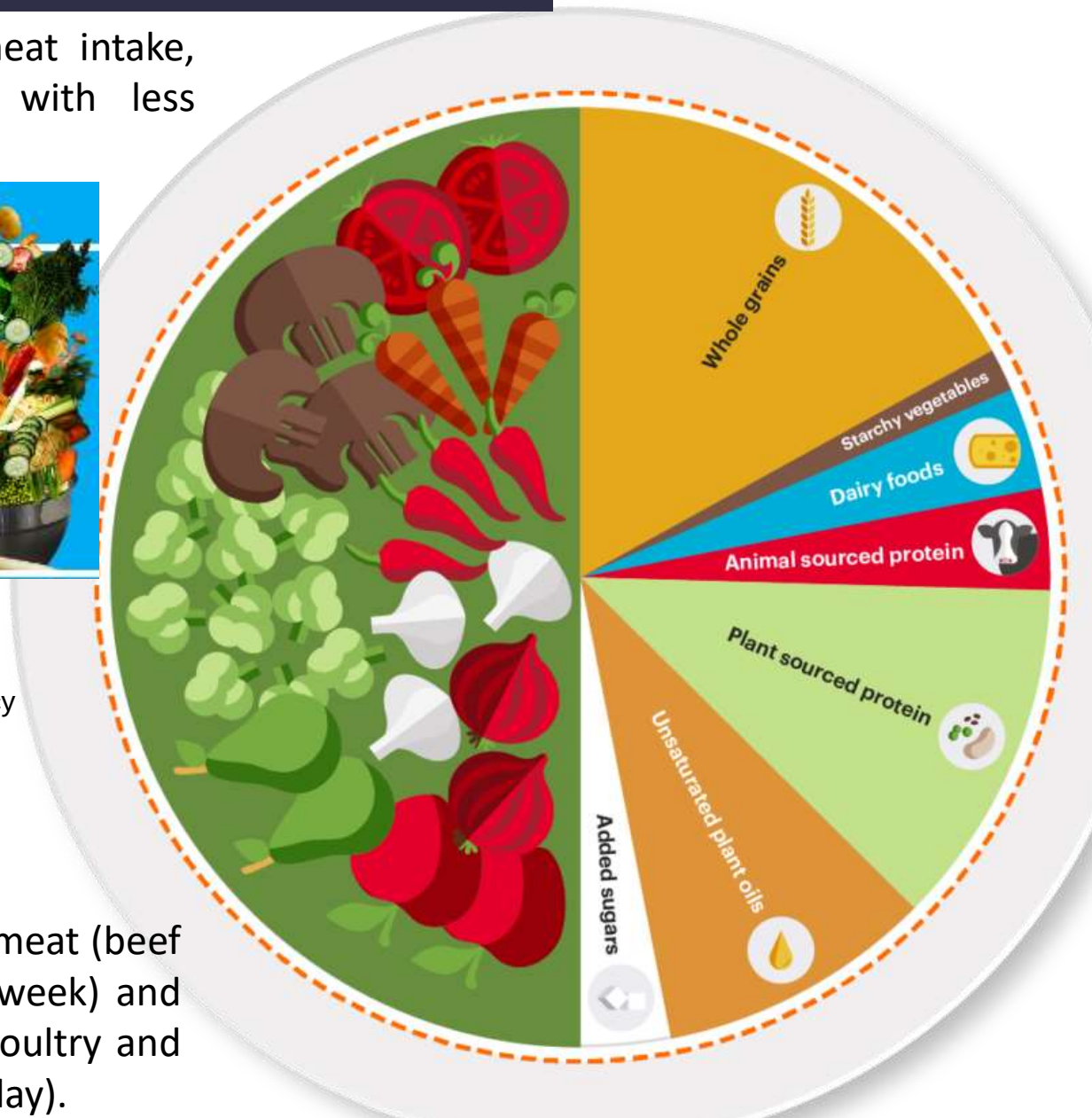
Healthy Calories



Best farm practices (lower emissions intensity): High efficiency

Plant-Rich Diet*
(reduced meat, not vegan)

... the sparing consumption of red meat (beef and pork; about one serving per week) and the limited consumption of fish, poultry and eggs (up to two servings each per day).



...4.3.2. Climate Change Mitigation

...4.3.2.3. Shifting to sustainable healthy diets or Dietary Shift

“Eat food. Not too much. Mostly plants.” Michael Pollan

- Healthier and more sustainable diets have multiple benefits, including benefits to human health and land use.
- *Rebalancing our diets* and reducing waste can prepare the ground for more sustainable farming practices. This includes supporting more diverse, nature-friendly agroecological practices, and ensuring that consumption doesn't lead to the loss of natural ecosystems overseas.
- *Rebalancing our diets* includes nutritional diversity for healthy diets and diversity of food culture. Such a shift would also score major wins for nature-protection and human health.
- Produce/use substitutes for the animal protein food/feed chains since these are seen as the principal source of biodiversity loss, climate change and land utilization by agricultural activities.
- As the global demand for meat continues to rise, protein diversification, in particular the scaling of alternative proteins, is becoming a critical component to sustainably reach global climate goals while also addressing the world's growing nutritional needs. Investors are increasingly integrating financially material environmental risks related to food system into their investment decision-making. At the same time, companies that diversify into new sources of climate-smart protein are drawing attention. Yet, the climate opportunity in alternative proteins remains largely untapped.
- Sustainability in agriculture, food labeling e.g. eco-positive brands and labels, diet changes, and policy can help create a food system that not only provides enough food, but keeps us and the planet healthy.
- For every penny spent on food and drink, a further penny is needed to pay for the environmental cleanup and health impacts of routine antibiotic usage. Cheap food is not good value for money.

...4.3.2. Climate Change Mitigation

...4.3.2.4. Reduction of CH₄ and N₂O emissions in agriculture

- Sustainable agricultural intensification can reduce ecosystem conversion and CH₄ and N₂O emissions.
- Mitigation options employed in these primarily concern the decrease of enteric methane emissions and manure methane and nitrous oxide emissions, although further reductions may be possible through mechanisms such as decreasing nitrous oxide emissions from the cultivation of animal feed.
- Reducing emissions from livestock and manure and shifting to renewable power in the food system.

...4.3.2. Climate Change Mitigation

...4.3.2.5. Reduction of food loss and waste (FLW)

Reduction of FLW is a critical climate strategy. Approaches to mitigate FLW impacts include methane abatement, climate-smart circular economy, and monetization using carbon credits. The beneficial measures on methane reductions are strengthening and expanding food cold chains/storages; consumer education campaigns; facilitating donation of unsold or excess food. Larger reductions could potentially be achieved with policies focused on preventing loss and waste of cattle products, as these have the largest associated methane emissions.

Addressing FLW is critical to food security, nutrition and climate change mitigation, with technology playing an important role. E.g. cold storage/chain could contribute to preserving highly perishable goods, such as vegetables, fruits, animal products, and fish, as well as efforts to improve food safety. Digital technology can facilitate “triple wins” of increasing access to safe and nutritious food, safeguarding farmer livelihoods and profits and reducing methane emissions by improving traceability across multiple agricultural supply chains.

‘Stop Food Waste’ Campaigns encourage to “Prevent, Inspire and Repurpose:” prevent food waste through proper storage and handling, inspire others to waste less, and repurpose food through circular economy efforts.

...4.3.2. Climate Change Mitigation

4.3.2.6. Roles of agriculture in climate change mitigation

- The agriculture sector provides products for which there are no alternatives, such as food, feed and ecosystem services. Agriculture is also a source of biomass that is used for power generation. To contribute to the green transition, biomass resources (i.e. organic material that can be used as fuel for transport, or to produce heat or electricity) should be sourced sustainably.
- Modern agriculture technologies should enable improved crop, water, and livestock management. Land use and forestry remove some GHGs. However, climate change is affecting most of these carbon sinks. Land is a carbon sink, absorbing more carbon from the atmosphere than it releases. This is why sustainable agriculture and food systems are key towards green objectives.
- Hence, there are many land-related climate change mitigation options that do not increase the competition for land. Many of these options have co-benefits for climate change adaptation.
- That means land ecosystems take up large amounts of carbon. Many land management options exist to both reduce the magnitude of emissions and enhance carbon uptake. **These options enhance crop productivity, soil nutrient status, microclimate or biodiversity, and thus, support adaptation to climate change.**

4.3.3. Climate Adaptation in Health and Nutrition

Human health will benefit from integrated mitigation and adaptation options that mainstream health into food, infrastructure, social protection, and water policies. Effective adaptation options exist to help protect human health and wellbeing, including: improving access to potable water, reducing exposure of water and sanitation systems to flooding, and Heat Health Action Plans that include early warning and response systems. Balanced and sustainable healthy diets and reduced food loss and waste contribute to nutrition, health, biodiversity and other environmental benefits.

Public health policies to improve nutrition, such as increasing the diversity of food sources in public procurement, health insurance, financial incentives, and awareness-raising campaigns, can potentially influence food demand, reduce food waste, reduce healthcare costs, contribute to lower GHG emissions and enhance adaptive capacity.

4.3.4. Climate responses and adaptation options in Society, Livelihoods, and Economy



The term response is used here in addition to adaptation because some responses, such as migration, relocation and resettlement may or may not be considered to be adaptation.

In 2022, there was record of 61 million new internally displaced persons worldwide. While conflict is one of the main causes, natural disasters, whose frequency and intensity are exacerbated by climate change, are responsible for more than half of these displacements. For instance, the India-Pakistan heatwave and floods in 2022 caused 30 billion dollars in economic losses as well as non-economic impacts, such as the displacement of people, health impacts, and pressure on healthcare facilities.

Fig. Climate responses and adaptation options in Society, Livelihoods, and Economy

...4.3.4. Climate responses and adaptation options in Society, Livelihoods, and Economy

The hungry season—a period of shortfalls in food and savings before the following harvest—is a common feature of life for rural families in many developing countries.

While the challenge of the hungry season has previously been met with interventions that increase income such as credit or incentives for migration, this modest budget intervention addresses one of the hungry season's root causes, which is lack of savings. Treating the hungry season as a savings problem can complement existing policies that seek to support families through shortages while also establishing a foundation to improve their future welfare.

Reallocate their savings across the year, reducing hunger while setting themselves up for higher future yields. A simple, inexpensive budgeting intervention stretched farmers' savings by roughly one extra month while increasing farm yields by 9 percent in the following year.

...4.3.4. Climate responses and adaptation options in Society, Livelihoods, and Economy

Climate change, for African countries, means vulnerability. It is synonymous with poverty, migration and displacement.

Migration is an important way to diversify agricultural-based livelihoods, particularly in sub-Saharan Africa.

Livelihood diversification is another risk management strategy that involves a process through which rural families diversify a portfolio of social support capabilities and activities to survive and improve their living standards.

Diversification to non-farm livelihood strategies enables households to enhance food security and increase income and agricultural production by securing capital. It also helps them to cope with environmental stresses. Diversifying to off-farm activities requires a different level of investment to offer higher returns.

Furthermore, they demand higher investment in terms of social, human, and financial capital to start up, making poor households less likely to be involved in high-return sectors, and diversification may not substantially help smallholder farmers get out of poverty.

The diversity of livelihood options should be available to individual level/scale such as farmers.

...4.3. Climate-resilient interventions

4.3.5. Regenerative agriculture and circularity in the agri-food systems

“Creating regenerative systems is not simply a technical, economic, ecological or social shift: it has to go hand-in-hand with an underlying shift in the way we think about ourselves, our relationships with each other and with life as a whole”

Daniel Christian Wahl, Author of Designing Regenerative Cultures (2016)

At perhaps the deepest level, a pervasive **cultural narrative of separation** leaves us disconnected from nature, from each other and from ourselves; with the result that broken relationships impair our systems on a global scale. The current failure of food systems to deliver food for human and planetary flourishing calls for innovative approaches that address the structural barriers to transformation and activate deeper levels of change.

To transform food systems, we must work not only on policy, research and project implementation, but also on the **inner drivers of individual, collective and institutional behaviours** E.g. Conscious farming /food systems/ and community development, revitalizing and strengthening traditional wisdom, conscious consumption. Consciousness practices include: nature connection, mindfulness (e.g. mindful eating and/through mindful farming), deep listening, emotional awareness, compassion trainings, (non-violent) communication and conflict resolution. Increased connection to ourselves, each other and nature can activate the **transformative qualities and skills** needed to support the transition to regenerative food systems. Consciousness and mental models are increasingly recognized as the key to unlock systems change, and regenerative agriculture calls for regenerative mindsets and cultures for more sustainability, regeneration and equity.

...4.3. Climate-resilient interventions

...4.3.5. Regenerative agriculture and circularity in the agri-food systems

Regenerative agriculture is an inclusive agroecosystems approach for conserving land and soil, biodiversity, and improving ecosystem services within farming systems. 'Regenerative agriculture' covers a wide spectrum of approaches, but put simply, it's a method of farming that rebuilds organic soil matter and restores biodiversity – while sequestering dangerous carbon emissions. Regenerative agriculture broadly encompasses both conservation agriculture and more sustainable agroforestry techniques.

Farmers plant a range of trees, plants and crops to keep vegetation in the ground, pulling the carbon out of the atmosphere and improving degraded soil. Grazing animals provide natural fertiliser, and farmers avoid soil disturbance (like ploughing) to keep carbon locked underground.

It focuses on the regeneration of living soil /improvement of soil fertility /, traps carbon emissions, improved micro hydrology /e.g. water retention/, and conserving biodiversity at all levels while enhancing inputs use efficiency and ecosystem services. The approach helps to achieve food and nutritional security by boosting yields generating income or extensive economic benefits, ecological sustainable options /e.g. mitigation, adaptation/ and social benefits.

Regenerative agriculture practices will also promote connections between farms and other local businesses, driving circular economic practices.

Regenerative/organic agriculture produces yields up to 40% higher during droughts, can earn farmers greater profits and releases 40% fewer carbon emissions than conventional agricultural practices. Crop diversification focused on cover crops, legumes and perennials returns more carbon to soil and sustain soil microbes that play key roles in carbon storage.

...4.3. Climate-resilient interventions

...4.3.5. Regenerative agriculture and circularity in the agri-food systems

Regenerative agriculture is more than sustainable — it allows for continued growth, something modern farming can't claim. Regenerative agriculture is an approach to farming that uses soil conservation as the entry point to regenerate and contribute to the delivery of food production and other ecosystem services. It is a framework rooted in indigenous farming practices developed from observing and responding to nature. Nature is more complex than we could ever begin to design, and working with it is the best for us. Genetically modified organism (GMO), pesticides, fertilizers, intensive animal farming, the whole spectrum of everyday practices that have repercussions on climate, nature and our health.

Regenerative agriculture can be broken down into 5 key principles:

- 1) reduced soil disturbance;**
- 2) retention of green cover;**
- 3) retention of living roots in the soil;**
- 4) crop diversification and**
- 5) integration of livestock into rotations.**

Regenerative agriculture has at its core the intention to build and maintain healthy soils or restore degraded soils, which enhances land productivity and produces more nutritious food with a lower environmental footprint.

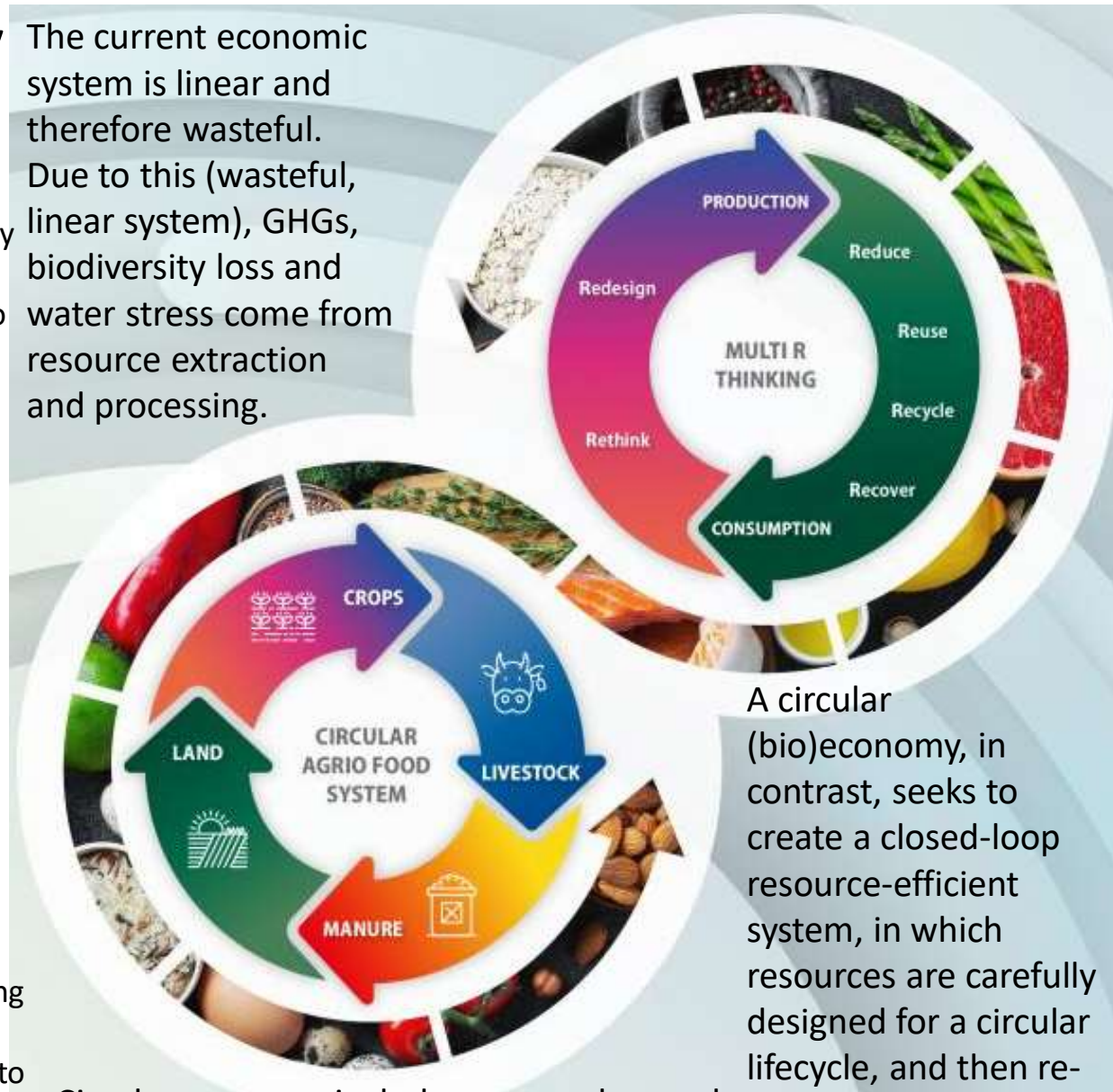
...4.3.5. Regenerative agriculture and circularity in the agri-food systems

Waste is incinerated, buried, or simply littered, wasting the potential utility and value of materials, and leading to ever more resource extraction.

A circular (bio)economy in the agri-food systems (circular agriculture) implies that agri-food products are generated in ways that regenerate nature, food is not lost and wasted, and all the resources are used productively. E.g. a circular economy approach to managing organic waste by converting it into regenerative agricultural inputs — insect-based protein for animal feed, organic fertilizer for farming and ecofuel.

The agri-food sector through practicing circular agriculture approaches has significant potential in the transition to low carbon and climate-friendly economy. Circularity holds of many benefits for people and planet.

The current economic system is linear and therefore wasteful. Due to this (wasteful, linear system), GHGs, biodiversity loss and water stress come from resource extraction and processing.



A circular (bio)economy, in contrast, seeks to create a closed-loop resource-efficient system, in which resources are carefully designed for a circular lifecycle, and then re-used, repaired, shared, or, in the last resort, recycled.

Circular economy includes approaches such as the 3R (reduce, reuse, and recycle) or 4R across the entire food system.

...4.3.5. Circularity in the agri-food systems

- Wasting food and non-food agricultural products, as well as inputs like plastics, puts unnecessary pressure on the environment.
- It means that we have wasted land and water resources, we have created pollution and we have emitted GHGs with no purpose.
- Bioeconomy promotes responsible production and consumption of renewable natural resources, contributing to climate action and protecting biodiversity
- Bioeconomy reduces waste and pollution, as it values and promotes responsible production and consumption of renewable natural resources, contributing to climate action, biodiversity conservation and ecosystems restoration.
- We must use existing biological resources more efficiently.
- For example, waste and residues along agrifood systems can be turned into beneficial products such as biomaterials, biochemicals, biopharmaceuticals, and bioenergy. In the bioeconomy, there is no such thing as waste.
- Identify the hotspots where losses and waste occur and to urgently address the inefficiencies and inequalities in global agrifood systems to make them more efficient, more inclusive, more resilient and more sustainable.

...4.3. Climate-resilient interventions

4.3.6. Integral soil improvement in climate change mitigation

As we integrate thinking around soil into climate, sustainability and food security strategies, momentum is building to bring soil onto the stage as part of land based solutions to climate change.

regenerative land management practices

Regenerative agriculture encompasses one or both of:

- regenerative (soil) practices (e.g. crop residue retention, cover crops or companion cropping, reduced tillage, use of organic manures or compost or crop residues, diverse rotations, improved fallow)
- whole system approaches (e.g. integrating livestock into rotations, integrated pest management, mob grazing or farm case studies).

Both can boost soil health or restore the complex soil biology that is key to long-term crop production, while also capturing more GHGs or increasing soil carbon in cropland and reducing nutrient runoff. More food *and* a healthier environment.



... 4.3. Climate-smart interventions

Fig.the collection of rabbit urine that is used in farming

...4.3. Climate-resilient interventions

Conditions that enable individual and collective actions

- Inclusive governance
- Diverse knowledges and values
- Finance and innovation
- Integration across sectors and time scales
- Ecosystem stewardship
- Synergies between climate and development actions
- Behavioural change supported by policy, infrastructure and socio-cultural factors

Centering mitigation and adaptation measures on human rights and effective participation is essential to tackling the root causes of climate change, avoiding harm to people and ecosystems, and dismantling structural inequalities and discrimination. Equity, social justice, climate justice, and human rights must be also at the core.

Climate justice is crucial because those who have contributed least to climate change are being disproportionately affected.

4.3.7. Holistic actions in climate-smart interventions

- Climate change is a global challenge that requires comprehensive and cross-sectoral action, including across agrifood systems. So, encourage holistic approaches centred on biodiversity and the interlinks with climate change.
- Hence, only holistic, integrated actions, springing from coordinated and collective efforts, will make our agrifood systems a central part of the global climate solution.
- A strategy on climate change must look beyond only food production by considering crops and livestock, forests, fisheries and aquaculture and related value chains, livelihoods, biodiversity and ecosystems in a holistic manner, as well as embracing the indispensable role of women, youth and Indigenous Peoples, as essential agents of change.

...4.3.7. Holistic actions in climate-smart interventions

- Community-based adaptation initiatives can be strengthened through the provision of training to local extension staff and local communities, including youth and women groups, for improved land, water and forest management.
- Integrating economic activities like beekeeping, honey production, fodder trees, nurseries for seedlings, etc with land restoration and water harvesting initiatives would not only create job opportunities, reduce poverty and generate a stable flow of income but also assure the sustainability of restoration projects over time and ultimately help to achieve the Greening objectives.
- Climate change-induced resource scarcity often leads to conflict and impedes efforts to tackle environmental degradation and achieve economic growth. We will have to work on pre-empting such conflicts through a concerted campaign and in collaboration with projects on the ground.